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(54) **STOCHASTIC TRAJECTORY PREDICTION USING SOCIAL GRAPH NETWORKS**

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(57) **ABSTRACT**

Systems, methods, apparatuses, and computer program products to provide stochastic trajectory prediction using social graph networks. An operation may comprise determining a first feature vector describing destination features of a first person depicted in an image, generating a directed graph for the image based on all people depicted in the image, determining, for the first person, a second feature vector based on the directed graph and the destination features, sampling a value of a latent variable from a learned prior distribution, the latent variable to correspond to a first time interval, and generating, based on the sampled value and the feature vectors by a hierarchical long short-term memory (LSTM) executing on a processor, an output vector comprising a direction of movement and a speed of the direction of movement of the first person at a second time interval, subsequent to the first time interval.

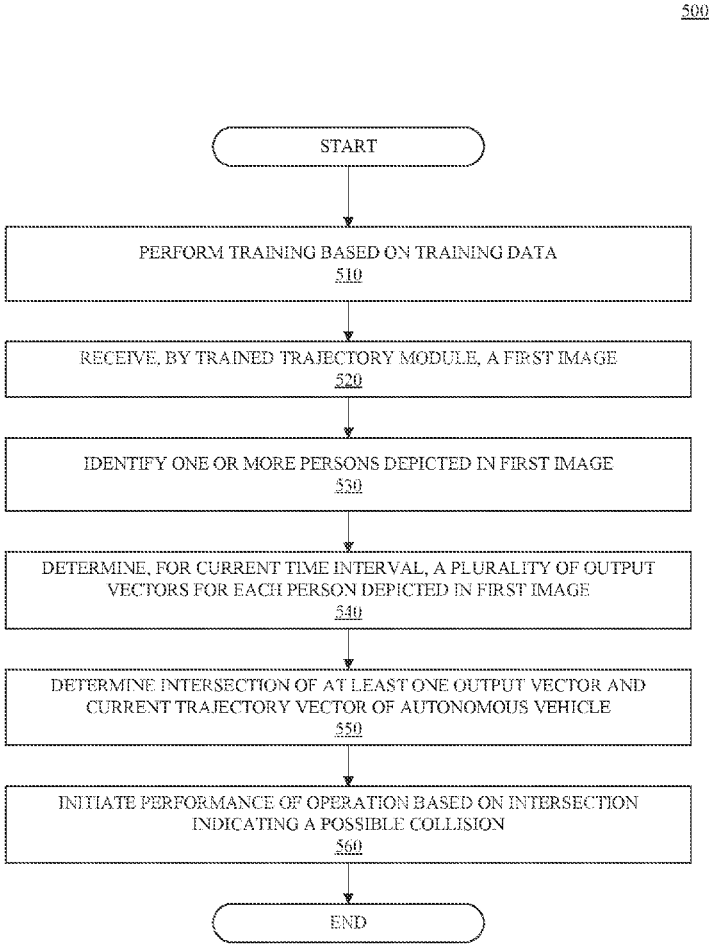
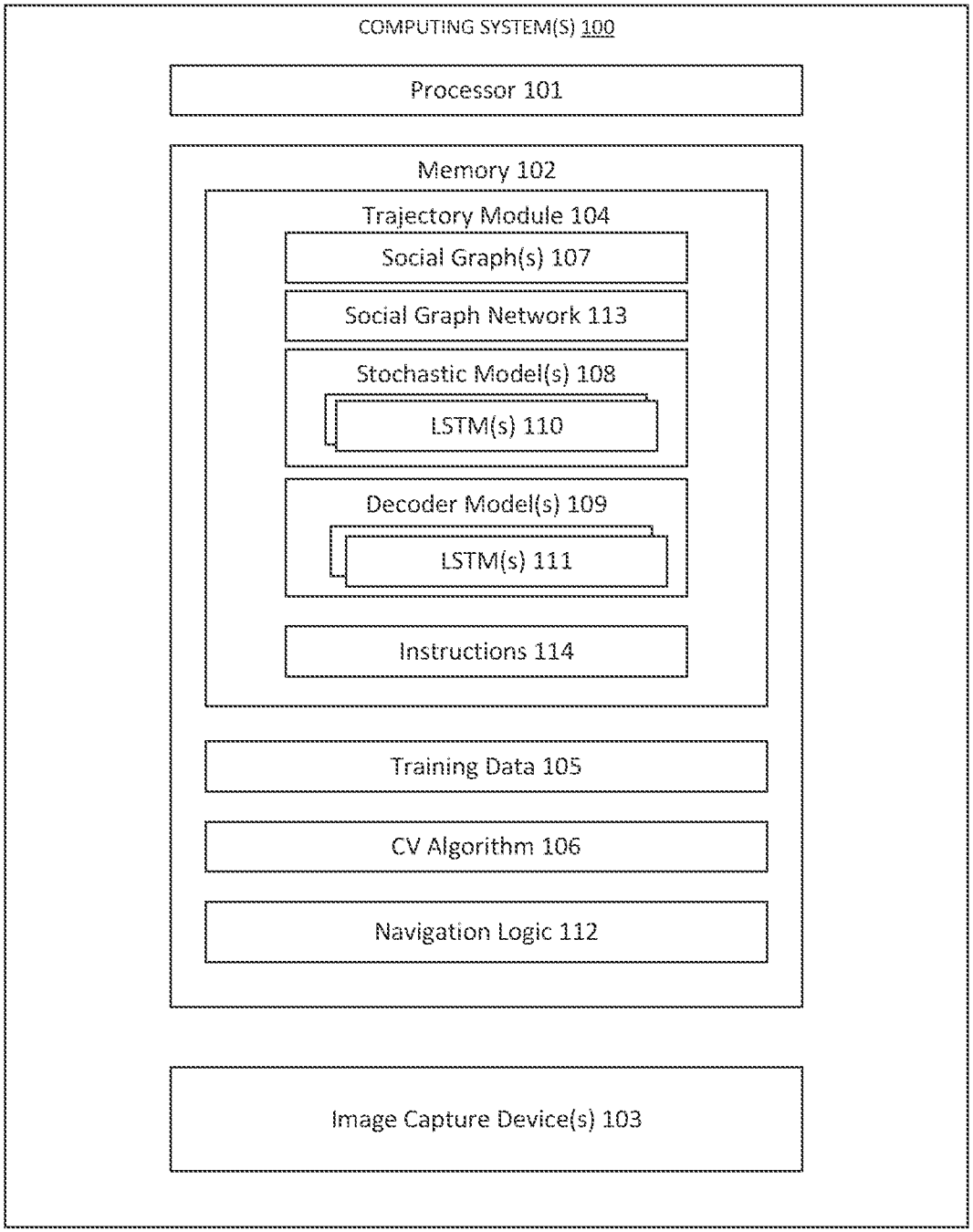


FIG. 1


COMPUTING SYSTEM(S) 100
Processor 101
Memory 102
Trajectory Module 104
Social Graph(s) 107
Social Graph Network 113
Stochastic Model(s) 108
LSTM(s) 110
Decoder Model(s) 109
LSTM(s) 111
Instructions 114
Training Data 105
CV Algorithm 106
Navigation Logic 112
Image Capture Device(s) 103

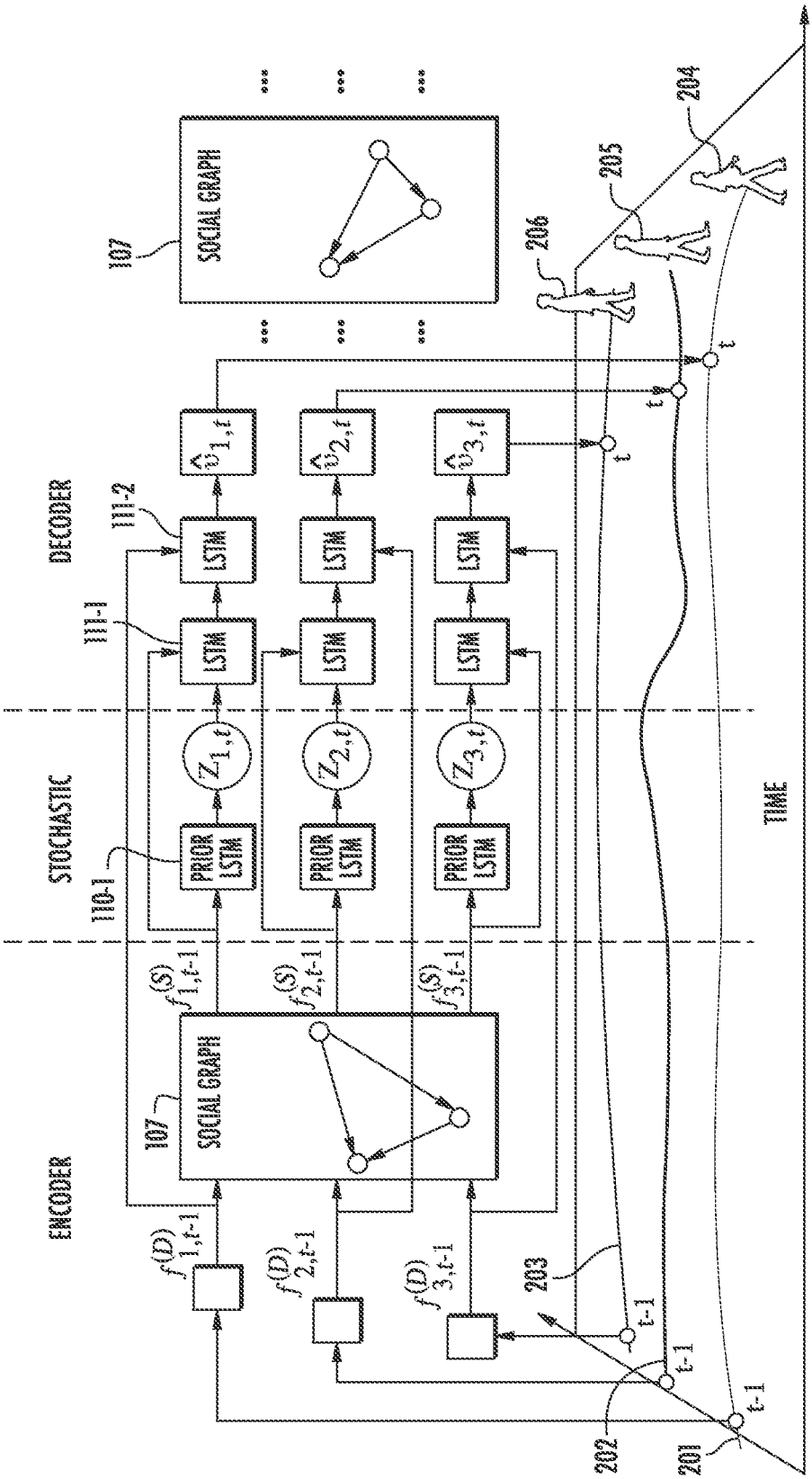

ENCODER
STOCHASTIC
DECODER
TIME
SOCIAL GRAPH
107
110-1
111-1
111-2
PRIOR LSTM
LSTM
203
202
201
204
205
206
200

FIG. 2

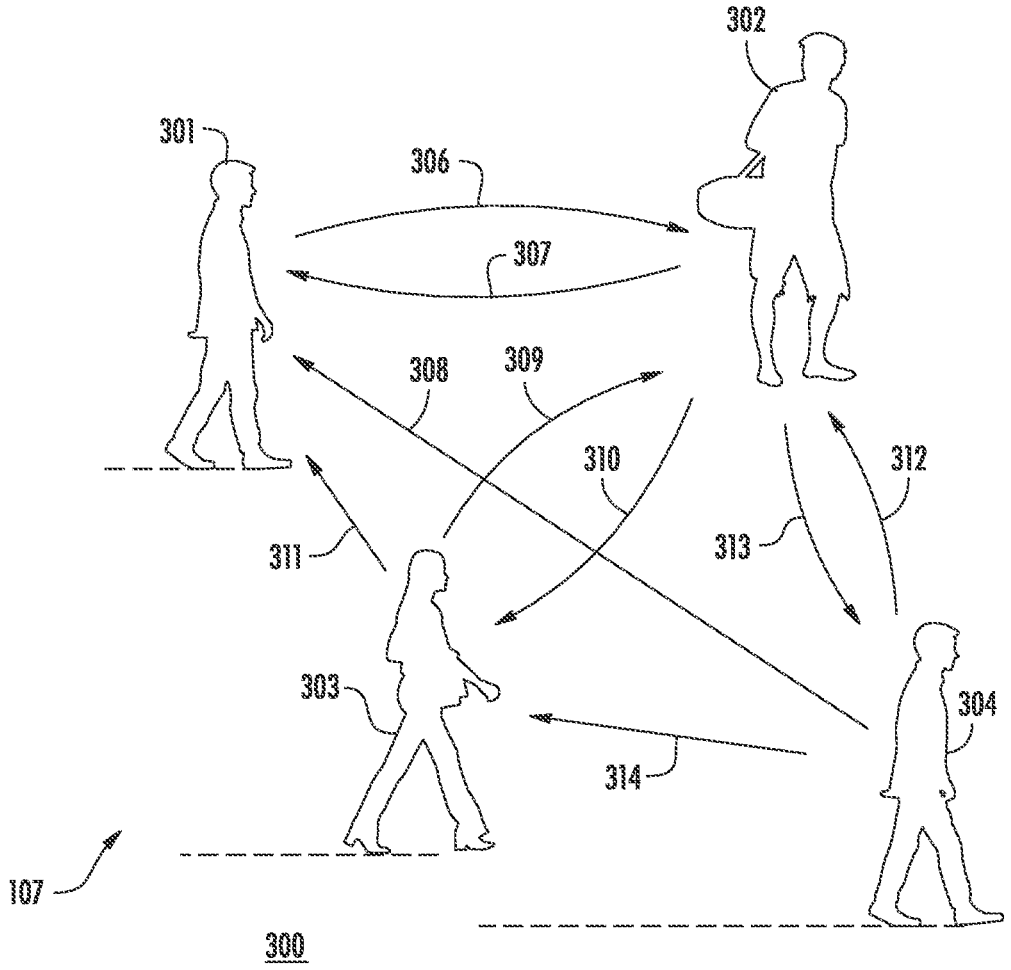


FIG. 3A

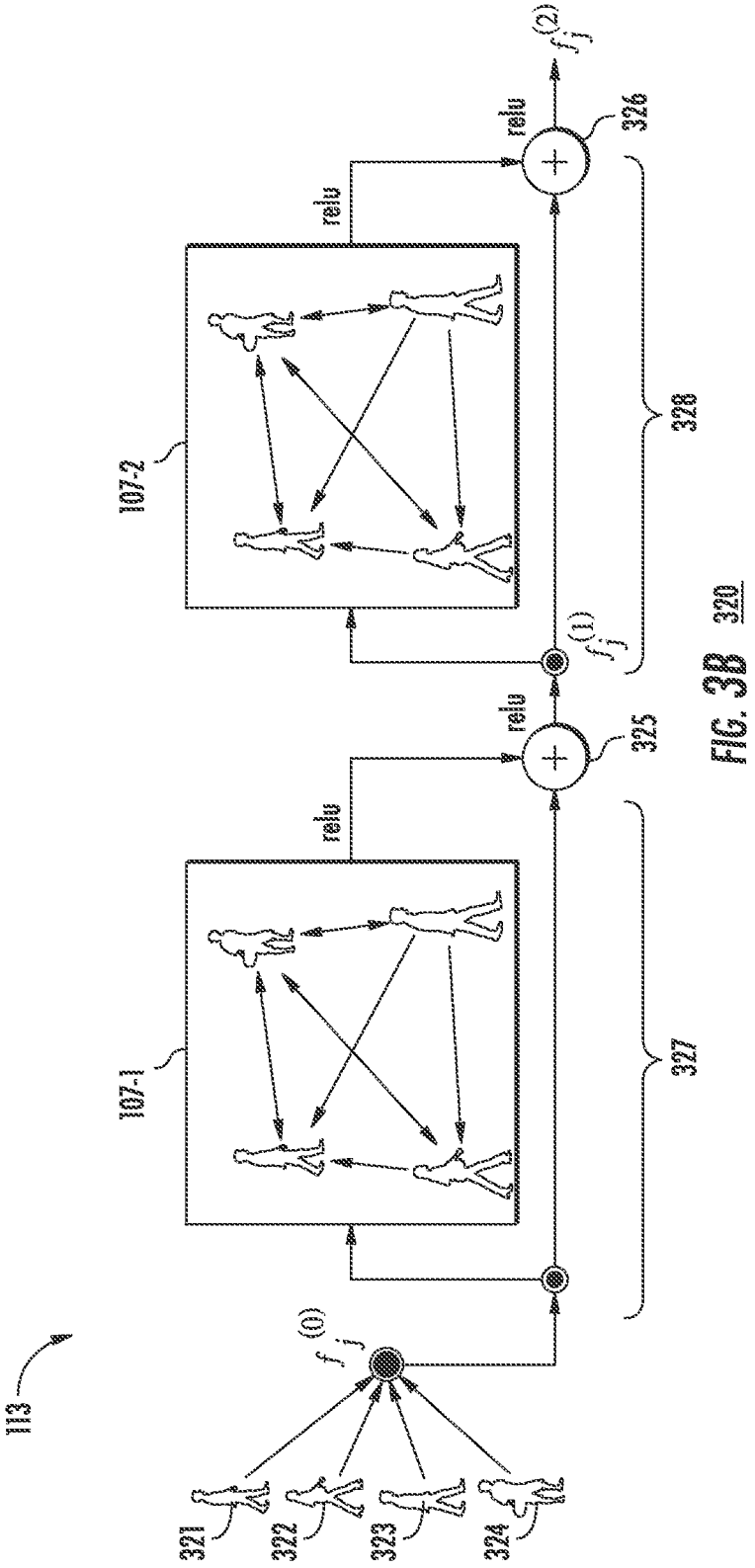

113
$f_j^{(0)}$
321
322
323
324
107-1
107-2
relu
relu
relu
relu
325
326
327
328
$f_j^{(1)}$
$f_j^{(2)}$
320

FIG. 3B

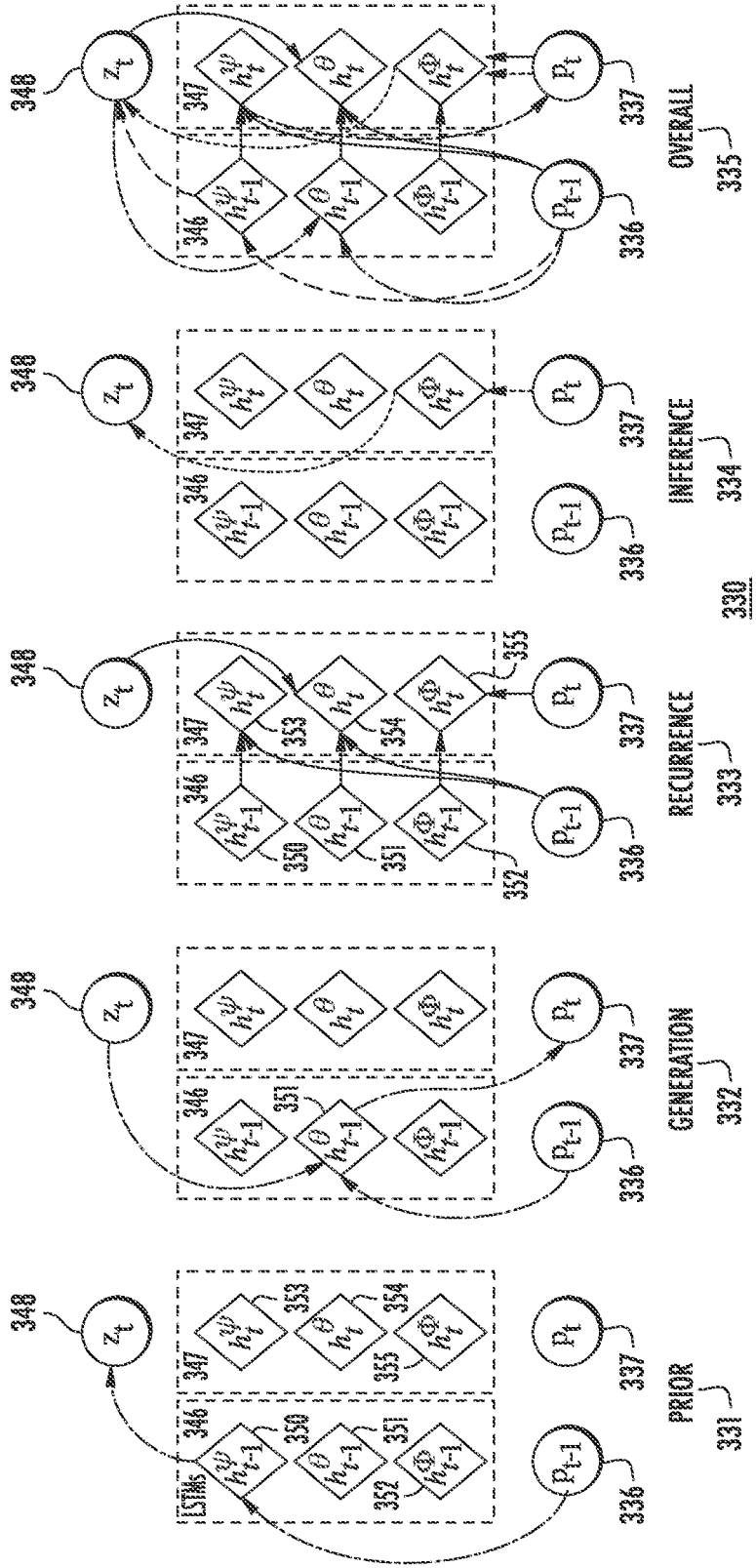

PRIOR
331
GENERATION
332
RECURRENCE
333
INFERENCE
334
OVERALL
335
330
LSTMs
346
347
348
336
337
350
351
352
353
354
355

FIG. 3C

FIG. 4

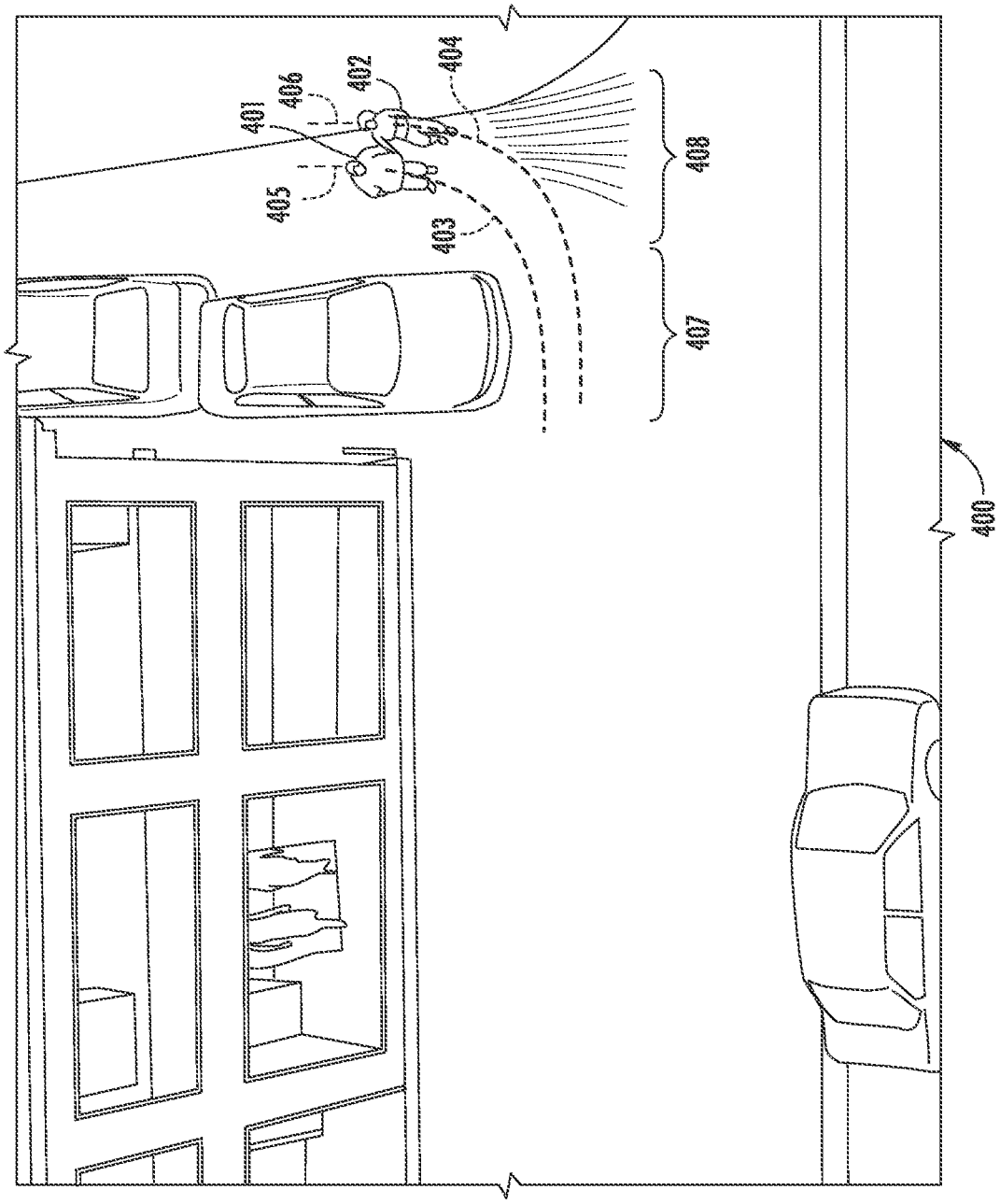


FIG. 5

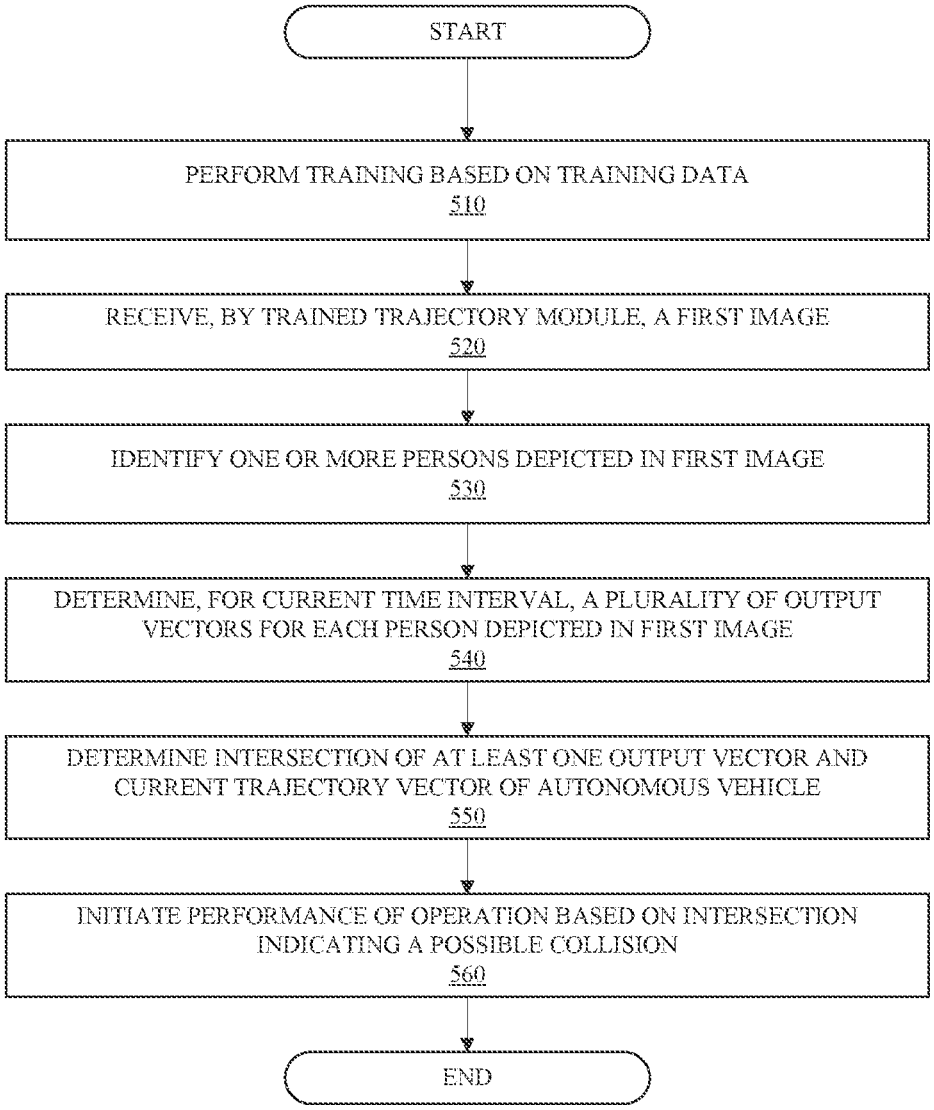

500
START
PERFORM TRAINING BASED ON TRAINING DATA
510
RECEIVE, BY TRAINED TRAJECTORY MODULE, A FIRST IMAGE
520
IDENTIFY ONE OR MORE PERSONS DEPICTED IN FIRST IMAGE
530
DETERMINE, FOR CURRENT TIME INTERVAL, A PLURALITY OF OUTPUT VECTORS FOR EACH PERSON DEPICTED IN FIRST IMAGE
540
DETERMINE INTERSECTION OF AT LEAST ONE OUTPUT VECTOR AND CURRENT TRAJECTORY VECTOR OF AUTONOMOUS VEHICLE
550
INITIATE PERFORMANCE OF OPERATION BASED ON INTERSECTION INDICATING A POSSIBLE COLLISION
560
END

FIG. 6

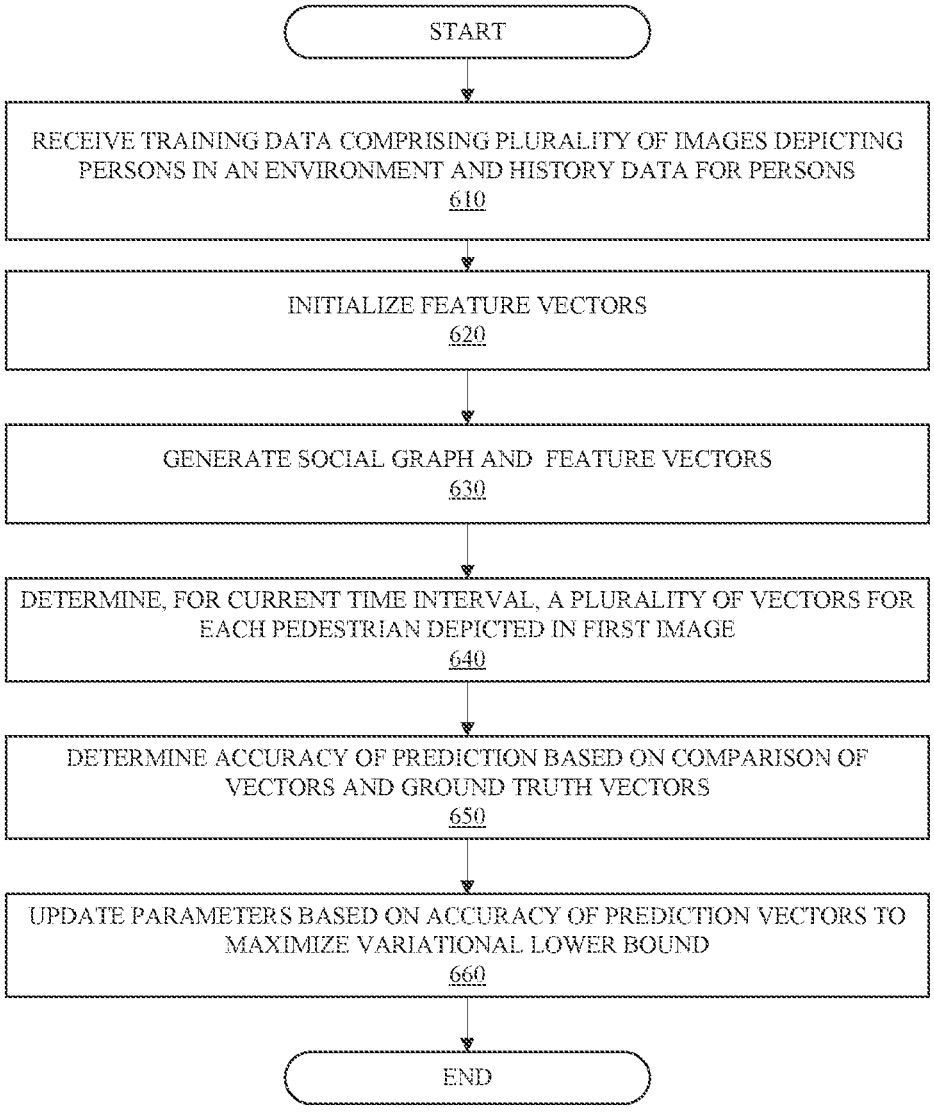

600
START
RECEIVE TRAINING DATA COMPRISING PLURALITY OF IMAGES DEPICTING PERSONS IN AN ENVIRONMENT AND HISTORY DATA FOR PERSONS
610
INITIALIZE FEATURE VECTORS
620
GENERATE SOCIAL GRAPH AND FEATURE VECTORS
630
DETERMINE, FOR CURRENT TIME INTERVAL, A PLURALITY OF VECTORS FOR EACH PEDESTRIAN DEPICTED IN FIRST IMAGE
640
DETERMINE ACCURACY OF PREDICTION BASED ON COMPARISON OF VECTORS AND GROUND TRUTH VECTORS
650
UPDATE PARAMETERS BASED ON ACCURACY OF PREDICTION VECTORS TO MAXIMIZE VARIATIONAL LOWER BOUND
660
END

FIG. 7

700

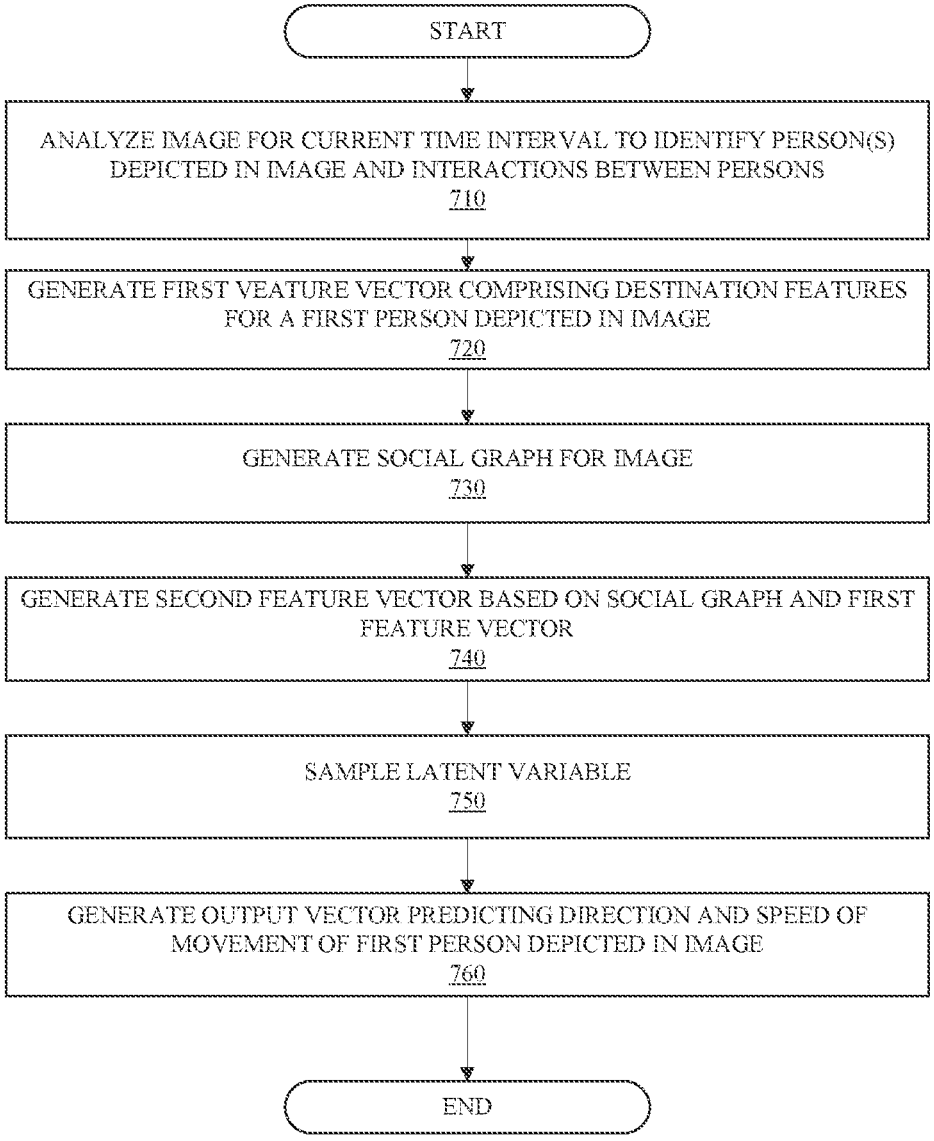

START
ANALYZE IMAGE FOR CURRENT TIME INTERVAL TO IDENTIFY PERSON(S) DEPICTED IN IMAGE AND INTERACTIONS BETWEEN PERSONS
710
GENERATE FIRST VEATURE VECTOR COMPRISING DESTINATION FEATURES FOR A FIRST PERSON DEPICTED IN IMAGE
720
GENERATE SOCIAL GRAPH FOR IMAGE
730
GENERATE SECOND FEATURE VECTOR BASED ON SOCIAL GRAPH AND FIRST FEATURE VECTOR
740
SAMPLE LATENT VARIABLE
750
GENERATE OUTPUT VECTOR PREDICTING DIRECTION AND SPEED OF MOVEMENT OF FIRST PERSON DEPICTED IN IMAGE
760
END

FIG. 8

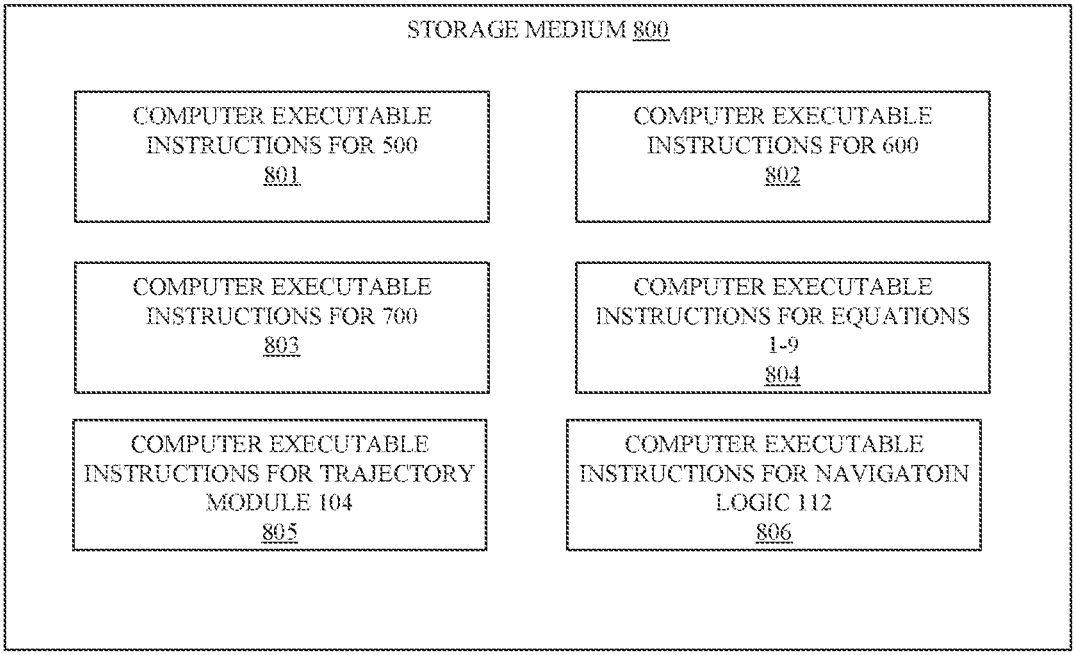
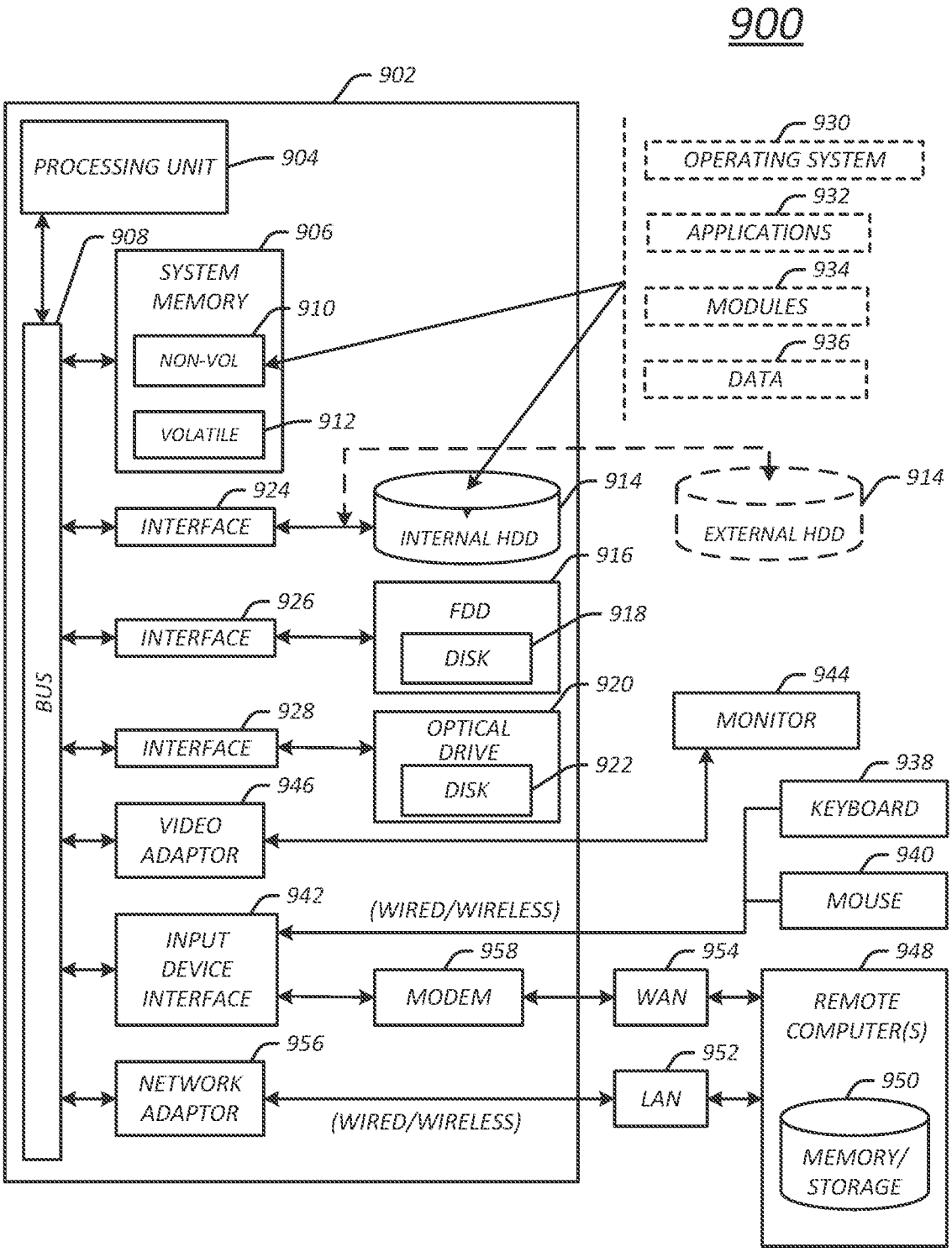

STORAGE MEDIUM 800
COMPUTER EXECUTABLE INSTRUCTIONS FOR 500 801
COMPUTER EXECUTABLE INSTRUCTIONS FOR 600 802
COMPUTER EXECUTABLE INSTRUCTIONS FOR 700 803
COMPUTER EXECUTABLE INSTRUCTIONS FOR EQUATIONS 1-9 804
COMPUTER EXECUTABLE INSTRUCTIONS FOR TRAJECTORY MODULE 104 805
COMPUTER EXECUTABLE INSTRUCTIONS FOR NAVIGATOIN LOGIC 112 806

FIG. 9


900
902
PROCESSING UNIT
904
908
906
SYSTEM MEMORY
910
NON-VOL
VOLATILE
912
930
OPERATING SYSTEM
932
APPLICATIONS
934
MODULES
936
DATA
924
INTERFACE
914
INTERNAL HDD
914
EXTERNAL HDD
916
926
INTERFACE
FDD
DISK
918
920
928
INTERFACE
OPTICAL DRIVE
DISK
922
944
MONITOR
938
KEYBOARD
946
VIDEO ADAPTOR
940
MOUSE
942
(WIRED/WIRELESS)
INPUT DEVICE INTERFACE
958
MODEM
954
WAN
948
REMOTE COMPUTER(S)
956
NETWORK ADAPTOR
952
LAN
950
(WIRED/WIRELESS)
MEMORY/ STORAGE
BUS

STOCHASTIC TRAJECTORY PREDICTION USING SOCIAL GRAPH NETWORKS

BACKGROUND

[0001] Programmatically predicting future locations of persons, vehicles, or other objects is difficult because of the uncertainty and/or ambiguity of any future motion. For example, a pedestrian approaching an intersection or leaving a building may move in any number of different directions at any given time. Furthermore, different external factors may impact the routes taken by the pedestrian. Therefore, it is difficult to accurately predict movement in different applications, such as autonomous vehicles, robots, drones, etc. Considering the amount of time and computing resources required for various computer vision algorithms (e.g., object detection, object tracking, etc.), the ability to more accurately predict future locations using fewer resources may provide significant improvements in collision prevention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. **1** illustrates an embodiment of a system.
[0003] FIG. **2** illustrates an example of stochastic trajectory prediction using social graph networks.
[0004] FIGS. **3**A-**3**C illustrate examples of stochastic trajectory prediction using social graph networks.
[0005] FIG. **4** illustrates an example image.
[0006] FIG. **5** illustrates embodiment of a first logic flow.
[0007] FIG. **6** illustrates an embodiment of a second logic flow.
[0008] FIG. **7** illustrates an embodiment of a third logic flow.
[0009] FIG. **8** illustrates an embodiment of a storage medium FIG. **9** illustrates an embodiment of a system.

DETAILED DESCRIPTION

[0010] Embodiments disclosed herein provide techniques to accurately predict possible future movement of persons depicted in one or more images using social graphs and a temporal stochastic method to model uncertainty of social interaction patterns. Generally, the social graph may model non-symmetric pairwise relationships and a social graph network may extract two types of features from the images and/or social graph. The two types of features may include individual destination-oriented features (e.g., features related to the destination of the person) and/or socially-aware features (e.g., features that describe human interactions). The social graph may comprise a directed graph that is updated at each of a plurality of time intervals given the location of persons depicted in the images and the velocity of any movement of the persons. Similarly, the temporal stochastic method to model uncertainty on social interactions between two or more persons depicted in the images may be updated at each time interval. Generally, at each time interval, the temporal stochastic method may sample a latent variable from a learned prior (that may vary across time) and use the sampled latent variable to generate diverse predictions. To generate all destination-oriented and/or social-plausible paths, the temporal stochastic method may leverage a hierarchical long short-term memory (LSTM) to progressively predict where the persons may move to next.
[0011] Advantageously, embodiments disclosed herein provide techniques to more accurately predict the movement of persons depicted in images. Doing so may improve the safety and reliability of different computing systems that predict where a person is moving. For example, using the techniques of the disclosure, a computing system may more accurately determine the future locations of one or more pedestrians depicted in an image. An autonomous vehicle may use the location data to determine that a future collision is likely to occur between the autonomous vehicle and one or more of the pedestrians. The autonomous vehicle may then perform an operation to avoid a collision with the pedestrian, e.g., by generating an alert that is outputted to the pedestrian (e.g., honking the horn of the autonomous vehicle) and/or changing the movement of the autonomous vehicle (e.g., slowing down, changing direction, and/or stopping). Embodiments are not limited in this context.

[0012] With general reference to notations and nomenclature used herein, one or more portions of the detailed description which follows may be presented in terms of program procedures executed on a computer or network of computers. These procedural descriptions and representations are used by those skilled in the art to most effectively convey the substances of their work to others skilled in the art. A procedure is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. These operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical, magnetic, or optical signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It proves convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. It should be noted, however, that these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to those quantities.

[0013] Further, these manipulations are often referred to in terms, such as adding or comparing, which are commonly associated with mental operations performed by a human operator. However, no such capability of a human operator is necessary, or desirable in most cases, in any of the operations described herein that form part of one or more embodiments. Rather, these operations are machine operations. Useful machines for performing operations of various embodiments include general purpose digital computers as selectively activated or configured by a computer program stored within that is written in accordance with the teachings herein, and/or include apparatus specially constructed for the required purpose. Various embodiments also relate to apparatus or systems for performing these operations. These apparatuses may be specially constructed for the required purpose or may include a general-purpose computer. The required structure for a variety of these machines will be apparent from the description given.

[0014] Reference is now made to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purpose of explanation, numerous specific details are set forth in order to provide a thorough understanding thereof. It may be evident, however, that the novel embodiments can be practiced without these specific details. In other instances, well known structures and devices are shown in block diagram form to facilitate a description thereof. The intention is to cover all modification, equivalents, and alternatives within the scope of the claims.

[0015] FIG. **1** illustrates an embodiment of a computing system **100** that provides stochastic trajectory prediction using social graph networks. The computing system **100** may be any type of computing system, such as a server, workstation, laptop, mobile device, or virtualized computing system. For example, the system **100** may be an embedded system such as a deep learning accelerator card, a processor with deep learning acceleration, a neural compute stick, or the like. In some examples, the system **100** comprises a System on a Chip (SoC) and, in other embodiments, the system **100** includes a printed circuit board or a chip package with two or more discrete components. The system **100** includes at least a processor **101**, a memory **102**, and one or more image capture devices **103**. The configuration of the computing system **100** depicted in FIG. **1** should not be considered limiting of the disclosure, as the disclosure is applicable to other configurations. The processor **101** is representative of any type of computer processor circuits, such as, central processing units, graphics processing units, or otherwise any processing unit. Further, one or more of the processors may include multiple processors, a multi-threaded processor, a multi-core processor (whether the multiple cores coexist on the same or separate dies), and/or a multi-processor architecture of some other variety by which multiple physically separate processors are in some way linked. Although one computing system **100** is depicted, multiple computing systems **100** may be communicably coupled via a communications network.

[0016] The memory **102** is representative of any type of information storage technology, including volatile technologies requiring the uninterrupted provision of electric power, and including technologies entailing the use of machine-readable storage media that may or may not be removable. Thus, the memory **102** may include any of a wide variety of types (or combination of types) of storage device, including without limitation, read-only memory (ROM), random-access memory (RAM), dynamic RAM (DRAM), Double-Data-Rate DRAM (DDR-DRAM), synchronous DRAM (SDRAM), static RAM (SRAM), programmable ROM (PROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), flash memory, polymer memory (e.g., ferroelectric polymer memory), ovonic memory, phase change or ferroelectric memory, silicon-oxide-nitride-oxide-silicon (SONOS) memory, magnetic or optical cards, one or more individual ferromagnetic disk drives, or a plurality of storage devices organized into one or more arrays (e.g., multiple ferromagnetic disk drives organized into a Redundant Array of Independent Disks array, or RAID array). It should be noted that although the memory **102** is depicted as a single block, the memory **102** may include multiple storage devices that may be based on differing storage technologies. Thus, for example, the memory **102** may represent a combination of an optical drive or flash memory card reader by which programs and/or data may be stored and conveyed on some form of machine-readable storage media, a ferromagnetic disk drive to store programs and/or data locally for a relatively extended period, and one or more volatile solid-state memory devices enabling relatively quick access to programs and/or data (e.g., SRAM or DRAM). It should also be noted that the memory **102** may be made up of multiple storage components based on identical storage technology, but which may be maintained separately as a result of specialization in use (e.g., some DRAM devices employed as a main storage while other DRAM devices employed as a distinct frame buffer of a graphics controller).

[0017] The image capture device is **103** representative of any type of device that captures image data depicting an environment. One example of an image capture device **103** includes a camera with an image sensor configured to capture digital images (e.g., color images, black and white images, grayscale images, x-ray images, infrared images, etc.) depicting an environment. The environment may be any real-world environment and/or a portion thereof, such as an intersection, street, sidewalk, waterway, airspace, room, hallway, floor of a building, etc. Often, the environment is crowded, and many people may be depicted in a given image. Although depicted as a component of the computing system **100**, the image capture device **103** may be external to the computing system **100**. In such an example, the image capture device **103** may be representative of one or more cameras (e.g., surveillance cameras) configured to capture images of an environment and provide the images to the computing system **100** (e.g., via a communications network) for processing as described in greater detail below. As another example, however, the surveillance camera may include the components of the system **100**, which may process the captured images as described in greater detail below.

[0018] In at least one embodiment, the computing system **100** is a component of (and/or communicably coupled to) an autonomous vehicle, drone, robot, or other device capable of partial and/or fully autonomous movement (collectively referred to as autonomous vehicles herein). In such examples, a navigation logic **112** may programmatically control at least a portion of the movement of the autonomous vehicle. Furthermore, the navigation logic **112** may receive signals from the trajectory module **104** based on processing of images captured by the image capture device **103**. For example, the trajectory module **104** may determine that a collision is likely to occur (e.g., beyond a threshold level of likelihood) with a pedestrian depicted in an image. In such an example, the navigation logic **112** may modify the movement of the autonomous vehicle (e.g., change direction of movement, change the speed of movement, stop movement, etc.). Similarly, the trajectory module **104** and/or the navigation logic **112** may output a warning signal (e.g., honking a horn of the autonomous vehicle, emitting light signals from the autonomous vehicle, etc.). More generally, regardless of the implementation, the trajectory module **104** may output warning signals that include audio signals, visual signals, and/or data signals. For example, when integrated into a surveillance camera, the trajectory module **104** may emit an audio signal via a speaker, emit light signals via a beacon, and/or transmit a data signal to the trajectory module **104** and/or navigation logic **112** of a nearby autonomous vehicle via a wireless network.

[0019] As shown, the memory **102** includes a trajectory module **104**, training data **105**, one or more computer vision (CV) algorithms **106**, and navigation logic **112**. Although depicted as residing in the memory **102**, the trajectory module **104**, training data **105**, CV algorithms **106**, and navigation logic **112** may be implemented as hardware, software, and/or a combination of hardware and software. For example, the processor **101** may execute the instructions **114** of the trajectory module **104** to perform the functionality of the trajectory module **104** described herein. Similarly, the processor **101** may execute instructions (not pictured) of the

CV algorithms **106** and/or navigation logic **102**. In some embodiments, the trajectory module **104** may be a component of the CV algorithms **106** and/or the navigation logic **112**.

[0020] The trajectory module **104** is generally configured to generate output reflecting predicted movement of objects depicted in one or more images, e.g., images captured by the image capture device **103** and/or images received from another source. While people are used as a reference example of objects herein, the disclosure is applicable to predicting the movement of other types of objects, such as autonomous vehicles, robots, animals, and the like. Generally, predicting the movement of humans is a challenging task for CV algorithms **106** (or other computing systems), as human movement may be affected by complex human social interaction behaviors and/or the random nature of human movement. For example, a person walking on a street may move to the left and/or move to the right to pass another person walking in front of them. Advantageously, embodiments disclosed herein address these issues by leveraging one or more social graphs **107** to capture the human interactions depicted in the images, a social network **113** to extract features from the images and/or social graph **107**, one or more stochastic models **108** to generate and sample latent variables conditioned on current states encoding both individual information and interaction information, and one or more decoder models **109** to predict the movement of persons depicted in the images. The types of human (or social) interactions may include, but are not limited to, a distance between two or more people depicted in image and/or whether one person is in view of another person in the image (e.g., based on whether vectors associated with two people intersect). The stochastic models **108**, decoder models **109**, and/or social network **113** may be any type of computing model, such as a neural network. The use of neural networks herein as reference examples of the stochastic models **108**, decoder models **109**, and/or social network **113** herein are not limiting of the disclosure, as the disclosure is applicable to all types of computing models that can be trained to perform the operations described herein.

[0021] For a given image (e.g., an image captured by the image capture device **103** and/or an image in the training data **105**), there may be N people depicted in the image, where N is a positive integer. The spatial location of the j-th person depicted in the image at time interval t may be denoted as $p_{j,t}=(x_{j,t}, y_{j,t})$, where x and y are the (x,y) coordinates in a 2-dimensional world-space and $0 \leq j \leq N$. Coordinates in the 2-dimensional world-space may be based on any measure of distance, such as meters, centimeters, inches, feet, etc. Doing so allows the trajectory module **104** to train and/or test based on absolute movement, whereas movement in pixel space for images may vary based on the parameters of the image capture device **103** (e.g., focal length, etc.). Therefore, the trajectory module **104** (or any component thereof) may convert image coordinates to the corresponding real-world coordinate (e.g., meters, centimeters) given known parameters of the image capture device **103**. Therefore, given previously observed images (e.g., images captured at prior time intervals depicting one or more of the N people) represented as T_{obs} and a trajectory history denoted by $\{p_{j,t}, j=1, \ldots N; t=1, \ldots, T_{obs}\}$, the trajectory module **104** may predict the trajectories of all N people, e.g., $\{p_{j,t}, j=1, \ldots N; t=T_{obs}+1, \ldots, T\}$. The trajectory history may be stored in the storage by the trajectory module **104** at each time interval.

[0022] The social graphs **107** are directed graphs that are generated at different time intervals (e.g., at 1 second intervals, 2 second intervals, etc.) based on the current location and velocity of people depicted in the image captured by the image capture device **103** at the corresponding time interval. Generally, the image capture device **103** may capture images at periodic time intervals, and the social graphs **107** may be generated to reflect the pairwise social relationships between people depicted in the images at the corresponding time interval. Based on an analysis of the captured images, the trajectory module **104** may identify persons in the image, determine the present location of the person, and update the trajectory history for each identified person (e.g., as metadata of the image and/or in a separate data store). The trajectory history may reflect the actual movement of each person at each time interval and may include a vector reflecting direction and/or velocity of movement at each time interval. The movement of each person at each time interval may be based on a respective image captured by the image capture device **103** depicting the person.

[0023] In one or more embodiments, a social graph **107** may be a directed graph G=(N;E;A), where N is a plurality of graph nodes, E is one or more graph edges connecting two nodes, and A is a non-symmetric adjacency matrix. Based on a given image (which may be analyzed by the CV algorithms **106** to identify persons, determine movement, determine that one person is in view of another person, identify interactions, the types of interactions, etc.), each pedestrian is assigned to a node ($n_j \in N$) in the social graph **107**, and an edge $e_{ij}=(n_i, n_j) \in E$ linking from i-th to j-th person exists when the adjacency matrix entry $a_{ij}=1$. Generally, at each time interval, the current position and speed direction of each person depicted in the corresponding image is used to determine whether another person is in the view of the person and generate the social graph **107** for the corresponding time interval. For example, a CV algorithm **106** and/or the trajectory module **104** may determine whether one or more rays emitted from a first person intersect with a second person in the image to determine whether the second person is in view of the first person at a given time interval. If the trajectory module **104** determines the person is in view, the trajectory module **104** may add an edge connecting the corresponding nodes in the social graph **107** for the time interval. However, if the first and second persons are no longer in view of each other at a later time interval, the social graph **107** for the later time interval will not include an edge connecting the first and second persons. Thus the social graph **107** is dynamically changed as the relative positions of people change across images.

[0024] The trajectory module **104** may then determine (or extract) two types of features describing the image and/or the N persons depicted in the image using the social graph network **113** based on the social graph **107** and/or the image. For example, processor **101** can execute instructions **114** of the trajectory module **104** to extract features from the images based on social graph network **113** and social graph **107**. A first type of feature may be individual destination features that may be denoted by $f^{(D)}$. The individual destination features may generally describe the destination of a given person, e.g., a target region in the image that serves as the destination of the person and/or any attribute thereof.

More generally, the individual features may further include the current position of each person in (x,y) coordinates, the trajectory history of the person at each time interval in (x,y) coordinates, etc. Furthermore, a second type of feature that encodes destination-oriented features and social features may be determined. The second type of features may be denoted as $f^{(S)}$, and may reflect the current position of the person, the trajectory history of the person, whether the person is communicating with another person, whether the person is walking with another person, etc. In one embodiment, the feature vectors $f^{(D)}$, $f^{(S)}$ may be vectors having 32 dimensions (e.g., values for 32 different features).

[0025] For the individual features $f^{(D)}$, the social graph network **113** may comprise a one-layer multi-layer perceptron (MLP) (e.g., a neural network) with a rectified linear unit (ReLU) to concatenate the (x,y) coordinates for the person $p_{j,t}$ and the velocity $v_{j,t}=p_{j,t}-p_{j,t-1}$ of the person as input. Stated differently, the individual features $f^{(D)}$ of person j at time interval t may be determined according to the following Equation 1:

$$f_{j,t}^{D}=W_d[p_{j,t},v_{j,t}]+b^d \quad \text{Equation 1.}$$

[0026] In Equation 1, (W^d, b^d) are weight and bias parameters of the social graph network **113**, respectively, and $v_{j,t}$ corresponds to the velocity of the person (determined over at least two time intervals) as demonstrated above. Therefore, Equation 1 extracts features based on the current position of each person, at least one prior location of each person, and the velocity of each person. In all remaining equations herein, subscripts t for time intervals are omitted for clarity.

[0027] As stated, the social graph network **113** may determine features $f^{(S)}$ that capture both individual destination features (e.g., $f^{(D)}$) as well as social interactions between two or more persons depicted in an image. In one embodiment, the social graph network **113** provides a ResNet-like structure that gradually learns pairwise social interactions in the residual branch by removing self-loops from the social graph **107**. Formally, the output of the L-th layer of the social graph network **113** to encode features for the j-th pedestrian may be denoted by Equation 2:

$$f_j^{L+1}=f_j^L+\left(W^g\left(\sum_{\forall i,a_{ij}=1}M_{ij}(W^i x_{ij}^L+b^i)\right)+b^g\right). \quad \text{Equation 2}$$

[0028] In Equation 2, $f_j^{(0)}=f_j^{(D)}$ at initialization, M_{ij} corresponds to a message passed from person i to person j in the social graph **107**, (W^i, b^i) denote weight and bias parameters for the input MLP with ReLU, and (W^g, b^g) denotes the weight and bias parameters for the global MLP with ReLU. The message may generally represent the first edge, e.g., that person i interacted with (and/or is in view of) person j in some way. The input x_{ij}^L to the social graph network **113** to compute vector $f^{(S)}$ may be denoted by Equation 3:

$$x_{ij}^L=[f_i^L,f_j^L,W^r\text{Polar}_{pj}(p_i)+b^r)] \quad \text{Equation 3.}$$

[0029] In Equation 3, $\text{Polar}_{pj}(p_i)$ corresponds to the local polar coordinates (p_i) having an origin of p_j. The polar coordinates may generally represent the distance between two people and/or the angle of a ray between two people. The message M_{ij} may be determined based on the following Equation 4:

$$M_{ij}^L=\sum_{\forall i,a_{ij}=1}\alpha_{ij}^L\cdot(x_{ij}^L\odot g_{ij}^L). \quad \text{Equation 4}$$

[0030] In Equation 4, α_{ij}^L is a scalar attention value applied to edge e_{ij} of the social graph **107**, g is a social gate, and $\odot$ is an element-wise multiplication operator. The scalar attention value α_{ij}^L may represent the degree of interaction, e.g., how close two people are to each other, the degree to which one person is in the view of another person, etc. Generally, Equation 4 serves as a filter to extract the relevant features at each time interval from Equation 3. Advantageously, the attention value α_{ij}^L measures the relative importance of each edge, while the social gate acts as an element-wise feature selector.

[0031] Equation 5 may be used by the social graph network **113** to determine the scalar attention value α_{ij}^l at each time interval:

$$\alpha_{ij}^l=\frac{\exp(Leaky\text{ReLU}(W^a x_{ij}^L))}{\sum_{\forall i,a_{ij}=1}\exp(Leaky\text{ReLU}(W^a x_{ij}^L))}. \quad \text{Equation 5}$$

[0032] Generally, the attention value α_{ij}^l reflects the degree of attention person i is giving to person j as reflected by the social graph **107** and the extracted features based on the weights applied to the location of each person. Stated differently, the attention value α_{ij}^l may be based on the distance between two people and/or the direction of movement of each person. Therefore, for example, Equation 5 may compute a larger attention value α_{ij}^l when two people are close to each other relative to an attention value α_{ij}^l computed when two people are far away from each other.

[0033] The social graph network **113** may compute the social gate g according to the following Equation 6:

$$g_{ij}^L=\text{sigmoid}(W^s x_{ij}^L+b^s) \quad \text{Equation 6.}$$

[0034] In example Equation 6, (W^s, b^s) correspond to the weight and bias parameters of the social gate, respectively. As shown, the social gate may use a sigmoid function to produce element-wise attention values. Generally, a feature vector may have a plurality of dimensions (e.g., a dimension for each feature of a plurality of features). However, it may be difficult to determine which dimensions, or elements, of the feature vector is important (or relevant). Therefore, the social graph network **113** may apply a filter (e.g., element-wise multiplication operation of the social gate) to the feature vector and generate the vector $f^{(S)}$. In at least one embodiment, the feature vector $f^{(S)}$ is an output of the last layer of the social graph network **113**.

[0035] The stochastic model **108** includes one or more LSTMs **110** to encode the vectors $f^{(D)}$, $f^{(S)}$ for each person and sample a latent variable φ for use as a value of a variable z_t added at each time interval and conditioned on $f^{(S)}$. An LSTM may be an artificial recurrent neural network (RNN) architecture that includes a cell, an input gate, an output gate and a forget gate (each not pictured). The cell may store values over time intervals while the gates regulate the flow

of information into and out of the cell. The stochastic model **108** may sample the latent variable according to the following Equations 7-8:

$$p_\psi(z_t|f_{<t}^S)=\mathrm{LSTM}_\psi(f_{t-1}^S) \quad \text{Equation 7.}$$

$$q_\phi(z_t|f_{\le t}^S)=\mathrm{LSTM}_\phi(f_t^S) \quad \text{Equation 8.}$$

[0036] Therefore, in the examples shown, the stochastic model **108** may include two LSTMs **110**, namely prior $\mathrm{LSTM}_\psi(f_{t-1}^S)$ in Equation 7, and posterior $\mathrm{LSTM}_\phi(f_t^S)$ in Equation 8. However, the stochastic model **108** may include any number of LSTMS **110**, and the use of two LSTMs should not be considered limiting of the disclosure. The prior $\mathrm{LSTM}_\psi(f_{t-1}^S)$ of Equation 7 may correspond to a Gaussian mean and variance, while the posterior $\mathrm{LSTM}_\phi(f_t^S)$ of Equation 8 may correspond to a Gaussian mean and variance. Generally, during training, the Gaussian distribution (e.g., mean and variance) of the prior $\mathrm{LSTM}_\psi(f_{t-1}^S)$ is refined to approximate the Gaussian distribution of the posterior $\mathrm{LSTM}_\phi(f_t^S)$. Once the distributions reach a threshold degree of similarity, the prior distribution of the prior $\mathrm{LSTM}_\psi(f_{t-1}^S)$ may replace the posterior distribution of the posterior $\mathrm{LSTM}_\phi(f_t^S)$. Therefore, the stochastic model **108** may sample the latent variable based on the Gaussian distribution of the posterior $\mathrm{LSTM}_\phi(f_t^S)$ during training and may sample the Gaussian distribution of the prior $\mathrm{LSTM}_\psi(f_{t-1}^S)$ during testing (or runtime, or inference) operations. The prior $\mathrm{LSTM}_\psi(f_{t-1}^S)$ may generally be learned based on past trajectory data of persons with recursive hidden states. The past trajectory data may include vectors describing the direction and speed of movement of the person at each time interval. The posterior $\mathrm{LSTM}_\phi(f_t^S)$ encodes scenes for the current time interval. As stated, the prior $\mathrm{LSTM}_\psi(f_{t-1}^S)$ is trained to approximate the posterior $\mathrm{LSTM}_\phi(f_t^S)$ to capture uncertain social interactions.

[0037] The decoder model **109** is generally configured to generate output to predict the movement of a given person depicted in an image at time interval t. In one embodiment, the decoder model **109** leverages hierarchical LSTMs **111** to progressively decode the feature vectors and predict the offset (e.g., an output vector) of the location of each person. The output generated by the decoder model **109** may take any form suitable to convey direction and/or speed of movement. For example, in one embodiment, the predicted movement may comprise a vector indicating a velocity and direction of movement (e.g., movement in the (x,y) direction at a velocity in meters per second). However, during training, the inputs to the decoder model **109** may comprise the ground truth data of the previous image (e.g., the actual movement of the person). The hierarchical LSTMs **111** may be a generation LSTM represented by LSTM_θ that stacks the two LSTMs with different inputs. The first LSTM **111-1** may receive social inputs (e.g., the feature vector $f^{(S)}$) to predict social reactions, and is combined with the second LSTM (e.g., an LSTM **111-2** for individual destination feature vector $f^{(D)}$) to generate socially-acceptable and destination-oriented trajectories. Equation 9 below may describe the operations performed by the decoder model **109**:

$$p_\theta(y_t|z_{\le t}f_{<t}^S f_{<t}^D)=\mathrm{LSTM}_\theta(z_t f_{t-1}^S f_{t-1}^D) \quad \text{Equation 9.}$$

[0038] In Equation 9, y_t corresponds to the output of the decoder model **109**, e.g., a vector specifying the predicted speed and direction of movement of a given person at a time interval based on the sampled latent variable z and the feature vectors $f^{(D)}$, $f^{(S)}$ for each person. As stated, the output vector y_t may be in any real-world unit of measure. In some embodiments, the decoder model **109** may compute a plurality of different estimated vectors y_t for each person for each person depicted in the image. In one such embodiment, the LSTM_θ of the decoder model **109** may correspond to Gaussian distribution with mean and variance. The decoder model **109** may sample the speed and/or direction of each person from this Gaussian distribution.

[0039] The trajectory module **104** (including any parameters of the stochastic model **108**, decoder model **109**, social graph network **113**, and/or LSTMs **110-111**) may be trained by maximizing the variational lower bound "ELBO". In one embodiment, the training of the trajectory module **104** may be based on the following Equation 10:

$$\mathcal{L}=\sum_{i=1}^{T}\left[E_{q_\phi(z_t|f_{\le t})}\log p_\theta(v_t \mid z_{\le t}, f_{<t}) - \beta D_{KL}(q_\phi(z_t \mid f_{\le t})\|p_\psi(z_t \mid f_{<t}))\right]. \quad \text{Equation 10}$$

[0040] Generally, the training uses the reparameterization trick to optimize Equation 10 with stochastic gradient descent during a backpropagation operation. The training may be based on a plurality of images in the training data **105** that are associated with trajectory data for each depicted person and ground-truth future locations indicating where each depicted person moved to next. In one embodiment, the trajectory module **104** converts trajectory coordinates of each person to world coordinates and interpolates the converted coordinates to sample the coordinate at periodic intervals (e.g., 0.5 seconds, 1 second, etc.). Therefore, for each training image, the trajectory module **104** analyzes each image and generates the social graph **107**. The social network **113** may then extract the feature vectors $f^{(D)}$, $f^{(S)}$, the stochastic model **108** may sample the latent variable, and the decoder model **109** may predict the next location of each person depicted in the image. The weights, biases, activations, and any other learnable parameters (e.g., of the models **108-109**, LSTMs **110-111**, social graph network **113**, etc.) may then be refined during training based on how close the predicted location for each person is to the ground-truth location for each person (e.g., on the accuracy of the predicted location generated by the decoder model **109**).

[0041] The accuracy may be based on any feasible metric, such as average displacement error (ADE) and/or final displacement error (FDE). The ADE may correspond to the averaged Euclidean distance between the ground truth and predicted location coordinates over all time intervals. The FDE may correspond to the Euclidean distance between the ground truth and the predicted coordinates in a final image of a scene (e.g., if a person is depicted in 5 images, the predicted location generated based on the fifth image). The value of hyperparameter β of Equation 10 may be based on a balance of reconstruction error and sample diversity. The left side of Equation 10 may be reduced to l_2 reconstruction loss between the predicted result and the ground-truth (e.g., the predicted location output of the decoder model **109** relative to the actual location reflected by the ground-truth of the training data **105**).

[0042] Once trained, the trajectory module **104** may more accurately predict the future movement of people depicted in images captured by the image capture device **103**. For example, if an image depicts 10 people in a scene, the

trajectory module **104** may predict the movement of each person. However, if the trajectory module **104** determines that the predicted location of one or more people may result in an unsafe situation (e.g., may result in a collision with an autonomous vehicle, robot, etc., based on one or more vectors generated by the trajectory module **104** intersecting with a vector corresponding to the trajectory of the autonomous vehicle), the trajectory module **104** may perform any number of operations. For example, the trajectory module **104** may provide the predicted location of the one or more people to the navigation logic **112**, which may modify the movement of the associated autonomous vehicle to avoid the collision. As another example, audio and/or visual alerts may be outputted by the trajectory module **104** to warn the persons of the possible danger. As another example, to determine whether the predicted location may result in a collision, the trajectory module **104** may determine whether a likelihood of collision exceeds a collision threshold. The likelihood of collision may be based on the predicted movement of the persons and/or vehicles. For example, if the vectors for one or more persons are determined to intersect with a vector for a vehicle, the trajectory module **104** may determine that the likelihood of collision exceeds the collision threshold. In some embodiments, the trajectory module **104** may compute a score reflecting the likelihood of collision.

[0043] FIG. **2** is a schematic **200** illustrating an example of stochastic trajectory prediction, according to one embodiment. As shown, the schematic **200** depicts example trajectory histories **201-203** for persons **204-206** depicted in one or more images. On the left side of FIG. **2**, the trajectory histories **201-203** may include locations of each person **204-206** at an example time interval t−1. Generally, the trajectory module **104** may then predict the location of each person **204-206** at time interval t, where time interval t is later in time than time interval t−1.

[0044] As shown, the feature vectors $f_{1,t-1}^{(D)}$, $f_{2,t-1}^{(D)}$, $f_{3,t-1}^{(D)}$ may be computed for each person **204-206**, respectively, e.g., based on Equation 1 above. As stated, these feature vectors may correspond to destination-based features, such as where each person **204-206** is traveling towards, the velocity of travel, prior history of movement, etc. Based on the extracted features $f_{1,t-1}^{(D)}$, $f_{2,t-1}^{(D)}$, $f_{3,t-1}^{(D)}$ and/or the analysis of each image, the social graph **107** may be generated. As stated, the social graph **107** represents each person identified in an image as a node. If two people are determined to interact in any way, an edge may connect the nodes representing the two people in the social graph **107**.

[0045] As further shown in FIG. **2**, the feature vectors $f_{1,t-1}^{(S)}$, $f_{2,t-1}^{(S)}$, $f_{3,t-1}^{(S)}$ may be computed for each person **204-206**, respectively, by the social graph network **113** based on the images, the feature vectors $f_{1,t-1}^{(D)}$, $f_{2,t-1}^{(D)}$, $f_{3,t-1}^{(D)}$ and/or the social graph **107**. Generally, the feature vectors $f_{1,t-1}^{(S)}$, $f_{2,t-1}^{(S)}$, $f_{3,t-1}^{(S)}$ may be computed according to Equations 2-6 above, and reflect destination-oriented as well as social features (e.g., whether each person is interacting with one or more other people).

[0046] For each person **204-206**, the stochastic model **108** may include an LSTM **110-1** to receive the corresponding feature vector $f^{(S)}$ as input and sample the latent variable z based on Equations 7-8 above. The LSTM **111-1** of the decoder model **109** may then receive the feature vector $f^{(S)}$ and the sampled latent variable z as input. The output of LSTM **111-1** may then be provided as input to the LSTM **111-2** of the decoder model **109** along with the feature vector $f^{(D)}$. The decoder model **109** may then compute vectors $\hat{v}_{1,t}$, $\hat{v}_{2,t}$, $\hat{v}_{3,t}$ reflecting the predicted movement of each person (e.g., speed of travel in the direction of the (x,y) coordinates). In one embodiment, the vectors are computed according to Equation 9 above. Generally, the vectors may include (x,y) coordinates indicating a direction of movement (e.g., (2,5) units in the x- and y-directions, respectively) and a velocity (e.g., n units per second, where n is any numerical value) of the direction movement of each person. These operations may be repeated any number of times to predict movement at future time intervals (e.g., t+1, t+2, . . . , t+n).

[0047] FIG. **3**A is a schematic **300** illustrating an example representation of a social graph **107**. As shown, FIG. **3**A depicts four example persons **301-304**. Each person **301-304** may be represented as a node in the social graph **107**. The edges **306-314** reflect that one of the persons **301-304** (e.g., person **304**) is in view of a different one of the persons **301-304** (e.g. person **303**), and the future path of person **304** may be affected by person **303**. More generally, the existence of an edge in the graph **107** is determined by pairwise positions. Therefore, if person A is ahead of (or in view of) person B, an edge in the graph **107** from person A to person B may be generated.

[0048] For example, edge **306** reflects that person **302** is in the view of person **301**, while edge **307** reflects that person **301** is in the view of person **302**. When the interaction is in one direction, only a single edge is generated in the social graph **107**. For example, edge **311** reflects that person **303** is paying attention to person **301**, while the absence of an edge from person **301** to person **303** indicates that person **301** is not interacting with person **303**.

[0049] FIG. **3**B is a schematic **320** illustrating an example social graph network **113**. Persons **321-324** in FIG. **3**B may correspond to the persons **301-304** depicted in FIG. **3**A. As shown, the social graph network **113** includes two stacked layers **327**, **328**, where the output of layer **328** is conditioned based on the output of layer **327**. For example, layer **327** may correspond to a previous time interval t=0, while layer **328** may correspond to a current time interval t=1. As shown, the input to layer **327** includes the feature vector $f_j^{(0)}$ (e.g., $f_j^{(S)}$ at time interval t=0) for each person **321-324**. The layers **327**, **328** learn pairwise interactions between the persons **321-324** in the residual branch of the ReLU by removing self-loops (e.g., a loop between the same person). Generally, the output for the layers **327**, **328** may be computed based on Equation 2 above (with additional operations performed according to Equations 3-6).

[0050] In the embodiment shown in FIG. **3**B, layer **327** of the social graph network **113** may leverage the social graph **107-1** of layer **327** to extract additional features for each person, which may be added to the input feature vector $f_j^{(0)}$ using the addition operator **325** to produce an output. The output of addition operator **325** may be the feature vector $f_j^{(1)}$, which may be provided as input to layer **328**. Features extracted from social graph **107-2** may then be added to feature vector $f_j^{(1)}$ using addition operator **326** to produce the output feature vector $f_j^{(2)}$.

[0051] FIG. **3**C is a schematic **330** depicting example operations performed by the trajectory module **104**, according to one embodiment. Prior phase **331** reflects computation of the conditional prior z_t **348** using Equation 7, where the prior is learned based on prior trajectories of each person with recursive hidden states. In FIG. **3**C, the "h" of LSTMs

350-355 reflect the hidden states of one or more of LSTMs **110-111**. As shown, the conditional prior is computed based on polar coordinates **336** of a person at time interval t−1 by LSTM group **356** (including LSTMs **350-352**). LSTM **350** may apply equation 7 to learn the conditional prior z_t **348**.

[0052] The generation phase **332** reflects a generation function used by the decoder model **109** to predict the location **337** of a person depicted in an image at time interval t based on Equation 9 and the location **336** at time t−1. As shown, the prior location **336** of the person and the sampled latent variable z_t from the prior **348** may be provided as input to LSTM **351**, which outputs the predicted location **337** of the person. As stated, the predicted location **337** may be a vector that indicates speed and direction of travel.

[0053] Recurrence phase **333** updates the recurrent hidden states h of the LSTMs **353-355** of LSTM group **347**. Generally, as shown, the hidden states h of the LSTMs **353-355** are updated based on the prior states h of the LSTMs **350-352**, the location coordinates **336** for time t−1, the sampled value from the conditional prior **348**, and the location coordinates **337** for time t.

[0054] Inference phase **334** reflects the inference of the latent variables based on Equation 8 above. As shown, the latent variable is inferred based on the location data **337** for time t. Overall phase **335** is a combination of phases **331-334** and may reflect the operations performed by the trajectory module **104** (and/or the components thereof).

[0055] FIG. **4** illustrates an example image **400**. The image **400** may be captured by an image capture device **103** and/or received by the trajectory module **104** from another source. The CV algorithms **106** may analyze the image **400** to identify persons **401**, **402** therein. As stated, destination feature vector $f_j^{(D)}$ may be computed for each person **401**, **402**. Similarly, a social graph **107** may be generated for the image **400**. The social graph **107** may assign persons **401**, **402** to respective nodes, and an edge may connect the nodes representing persons **401**, **402**. For example, the edge may associate persons **401**, **402** based on one or more of: that persons **401**, **402** are walking side-by-side, that persons **401**, **402** are in view of each other, that persons **401**, **402** are holding hands, and/or the ground truth trajectories **405**, **406** of persons **401**, **402**. The ground truth trajectories **405**, **406** may correspond to the actual paths of the persons **401**, **402** at prior time intervals. The social network **113** may then extract the feature vector $f_j^{(S)}$ for each person **401**, **402**, and the stochastic model **108** may sample a value for the latent variable z_t from the learned prior.

[0056] The decoder model **109** may then compute one or more predicted trajectories for each person **401**, **402**. As shown, the decoder model **109** may determine a plurality of trajectories **407**, **408**, for persons **401**, **402**, respectively. Generally, the trajectories **407**, **408** reflect the estimated future travel for each person **401**, **402**. However, the most likely trajectories **403**, **404**, may be returned as the most probable trajectories, e.g., by sampling the Gaussian distribution computed using equation 9 above. As shown, because of the interactions between persons **401**, **402**, the decoder model **109** computes related trajectories **403**, **404**, reflecting that the persons are likely to continue traveling together.

[0057] FIG. **5** illustrates an embodiment of a logic flow **500**. The logic flow **500** may be representative of some or all of the operations executed by one or more embodiments described herein. For example, the logic flow **500** may be representative of some or all of the operations to provide stochastic trajectory prediction using social graph networks. Embodiments are not limited in this context.

[0058] As shown, at block **510**, the training of the trajectory module **104** and/or the components thereof is performed based on the training data **105**. As stated, the training data **105** includes images depicting people moving in an environment. Each image of the training data **105** may include labels (or other types of metadata) indicating, for each person in the image, ground truth data reflecting the actual movement of the person (e.g., a vector comprising direction and speed) subsequent to the captured image (e.g., where the person actually moved at time t=1 subsequent to an image depicting the person at time t=0). At block **520**, the trained trajectory module **104** may receive a first image captured by an image capture device **103**. For example, a stationary surveillance camera may capture images of an environment at periodic time intervals, and the first image received at block **520** may correspond to one of the captured images of the environment.

[0059] At block **530**, the trajectory module **104** and/or a CV algorithm **106** may identify one or more persons depicted in the first image. For example, the persons may be detected using human detection, tracking, identification algorithms, etc. More generally, any feasible technique may be used to identify persons in a given image. At block **540**, the decoder model **109** may compute a plurality of vectors for each person, where each vector corresponds to a predicted speed and direction of movement of each person. In at least one embodiment, the decoder model **109** may return a first vector of the plurality of vectors as the most likely course of movement the person will take. In one embodiment, the first vector is returned by sampling the learned prior distribution generated based on Equation 9.

[0060] At block **550**, the trajectory module **104** may determine that one or more of the output vectors computed at block **540** intersects with a trajectory vector reflecting the movement of an autonomous vehicle. If the vectors intersect at one or more points in the environment, the trajectory module **104** may determine that a collision may occur between the person associated with the output vector and the autonomous vehicle. In some embodiments, the trajectory module **104** may determine that a likelihood of the collision exceeds a collision threshold, e.g., by computing a score reflecting the likelihood of collision, where the score is based on whether the vectors intersect. In one embodiment, the autonomous vehicle may be depicted in the first image. In such an example, the trajectory module **104** may determine the predicted movement of the autonomous vehicle based on previous images depicting the autonomous vehicle, the direction of travel of the autonomous vehicle, etc. In other embodiments, the computing system **100** may be a component of the autonomous vehicle. In such embodiments, navigation logic **112** may provide the predicted movement of the autonomous vehicle. The trajectory module **104** and/or the navigation logic **112** may then compare the vectors computed at block **540** to the projected movement of the autonomous vehicle to determine whether the vectors may intersect.

[0061] At block **560**, a predefined operation may be performed based on a determination that a collision may occur between an autonomous vehicle and one or more persons depicted in the first image. For example, if a first vector for a first person reflecting the most likely movement of the first person indicates a collision may occur with a first autono-

mous vehicle, an alert may be generated to warn the first person and/or the navigation logic **112** of the first autonomous vehicle. Stated differently, the predefined operation may be performed based on a determination that the likelihood of collision exceeds a collision threshold. Doing so may allow the navigation logic **112** to alter the movement of the autonomous vehicle to avoid the collision (e.g., slow down, stop, turn, etc.).

[0062] FIG. **6** illustrates an embodiment of a logic flow **600**. The logic flow **600** may be representative of some or all of the operations executed by one or more embodiments described herein. For example, the logic flow **600** may be representative of some or all of the operations to train the trajectory module **104**. Embodiments are not limited in this context.

[0063] As shown, at block **610**, training data **105** may be received by the trajectory module **104**. The training data **105** may comprise a plurality of images of an environment captured at different time intervals. The images may depict one or more persons. The training data **105** may further include ground-truth data reflecting the actual movement of the depicted persons (e.g., vectors indicating direction and speed of movement). At block **620**, the feature vectors $f^{(D)}$, $f^{(S)}$ may be initialized for each person depicted in a first image. At block **630**, the feature vectors $f^{(D)}$, $f^{(S)}$ and the social graph **107** are computed according to Equations 1-6 above. The feature vectors may be for each person depicted in the first image from the training data **105**. The feature vectors may reflect a destination of each person, which other persons are in view of a given person, and/or the distance between two people. The social graph **107** is generated for the first image. Furthermore, during training, Equations 7-8 above may be applied to learn the conditional prior distribution based on the posterior prior distribution and allow the stochastic model **108** to sample values therefrom. As stated, during training, values are sampled from the posterior distribution to learn the prior distribution.

[0064] At block **650**, the decoder model **109** applies Equation 9 to compute a plurality of vectors for each person depicted in the image. Each vector may generally describe a predicted direction and speed of movement of the person. The vectors may describe the direction in (x,y) coordinates and the speed of movement in any units and/or format (e.g., meters per second, feet per second, etc.). At block **660**, the accuracy of the vectors computed at block **650** is determined, e.g., based on a comparison of the vectors computed at block **650** to ground-truth movement for each person. As stated, the accuracy may be based on the average displacement error and/or final displacement error. Generally, for example, if a first vector computed at block **650** indicates a person will move to the left, but the ground truth data indicates the person actually moved to the right, the prediction computed at block **650** may have a low accuracy. Therefore, at block **660**, a backpropagation operation is performed to update the parameters of the trajectory module **104** (e.g., the social graph network **113**, the stochastic model **108**, the decoder model **109**, and/or the LSTMs **110**, **111**). Generally, Equation 10 above may be used to maximize the variational lower bound of Equation 10 using stochastic gradient descent. Although blocks **610-660** may be discussed with reference to a single image and/or a single person depicted in each image, the blocks **610-660** may be repeated any number of times on any number of images in the training data **105**. Doing so allows the trajectory module **104** to generate more accurate predictions over time.

[0065] FIG. **7** illustrates an embodiment of a logic flow **700**. The logic flow **700** may be representative of some or all of the operations executed by one or more embodiments described herein. For example, the logic flow **700** may be representative of some or all of the operations to provide stochastic trajectory prediction using social graph networks. Embodiments are not limited in this context.

[0066] As shown, at block **710**, an image may be analyzed by one or more CV algorithms **106** to identify one or more persons depicted in the image and/or interactions between at least two people depicted in the image. The image may be captured by the image capture device **103** and/or received from another source. The interactions may include whether one person is in view of another person, the distance between two people, communications, physical contact, one person avoiding another person, and the like. At block **720**, a first feature vector, namely a destination feature vector $f^{(D)}$ comprising one or more destination-based features is generated according to Equation 1 for each person identified in the image by a neural network (e.g., the social network **113** of the trajectory module **104**). As stated, the neural network may include the one-layer MLP with ReLU to concatenate the (x,y) coordinates of the person and the velocity of the person as input. The velocity may be based on a difference of the coordinates of the person at a current time interval and the coordinates of the person at a previous time interval (e.g., the preceding time interval). At block **730**, a social graph **107** is generated for the image. The social graph **107** is a directed graph representing each person in the image as a node. If two or more people are determined to engage in social interactions based on the analysis of the image at block **710**, an edge may connect the two people engaging in social interactions as described above. For example, if the distance between two people is below a threshold, a social interaction may be determined (and the attention value may be based on the distance). Similarly, if one person is in view of another person, an edge may connect the nodes representing the two people in the social graph **107**, and the attention value may reflect the degree to which the person is in view of the other person.

[0067] At block **740**, a second feature vector, namely the vector $f^{(S)}$ is computed according to Equations 2-6 for each person depicted in the image. The social graph network **113** may compute the second feature vector $f^{(S)}$ based on the first feature vector $f^{(D)}$ and/or the social graph **107**. The output of the last layer of the social graph network **113** may comprise the vector $f^{(S)}$. At block **750**, a value for the latent variable z is sampled from the learned prior distribution. At block **760**, the decoder model **109** may compute one or more output vectors reflecting the predicted direction of movement and a predicted speed of the direction of movement of each person depicted in the image. The decoder model **109** may then store the generated vectors for later use.

[0068] FIG. **8** illustrates an embodiment of a storage medium **800**. Storage medium **800** may comprise any non-transitory computer-readable storage medium or machine-readable storage medium, such as an optical, magnetic, or semiconductor storage medium. In various embodiments, storage medium **800** may comprise an article of manufacture. In some embodiments, storage medium **800** may store computer-executable instructions, such as computer-executable instructions to implement one or more of logic flows or

operations described herein, such as instructions **801**, **802**, **803** for logic flows **500**, **600**, **700** of FIGS. **5-7**, respectively. The storage medium **800** may further store computer-executable instructions **804** for Equations 1-9 described above, and computer-executable instructions **805** for the trajectory module **104** and components thereof (e.g., the social graph network **113**, the social graph **107**, the stochastic model **108**, the decoder model **109**, the instructions **114**, and/or the LSTMs **110**, **111**). The storage medium **800** may further store computer-executable instructions **806** for the navigation logic **112**. The processor **101** may execute any of the instructions **801-806**. Examples of a computer-readable storage medium or machine-readable storage medium may include any tangible media capable of storing electronic data, including volatile memory or non-volatile memory, removable or non-removable memory, erasable or non-erasable memory, writeable or re-writeable memory, and so forth. Examples of computer-executable instructions may include any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, object-oriented code, visual code, and the like. The embodiments are not limited in this context.

[0069] FIG. **9** illustrates an embodiment of an exemplary computing architecture **900** that may be suitable for implementing various embodiments as previously described. In various embodiments, the computing architecture **900** may comprise or be implemented as part of an electronic device. In some embodiments, the computing architecture **900** may be representative, for example, of a computer system that implements one or more components of the system **100**. The embodiments are not limited in this context. More generally, the computing architecture **900** is configured to implement all logic, systems, logic flows, methods, apparatuses, and functionality described herein and with reference to FIGS. **1-8**.

[0070] As used in this application, the terms "system" and "component" and "module" are intended to refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution, examples of which are provided by the exemplary computing architecture **900**. For example, a component can be, but is not limited to being, a process running on a processor, a processor, a hard disk drive, multiple storage drives (of optical and/or magnetic storage medium), an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a server and the server can be a component. One or more components can reside within a process and/or thread of execution, and a component can be localized on one computer and/or distributed between two or more computers. Further, components may be communicatively coupled to each other by various types of communications media to coordinate operations. The coordination may involve the uni-directional or bi-directional exchange of information. For instance, the components may communicate information in the form of signals communicated over the communications media.

[0071] The information can be implemented as signals allocated to various signal lines. In such allocations, each message is a signal. Further embodiments, however, may alternatively employ data messages. Such data messages may be sent across various connections. Exemplary connections include parallel interfaces, serial interfaces, and bus interfaces.

[0072] The computing architecture **900** includes various common computing elements, such as one or more processors, multi-core processors, co-processors, memory units, chipsets, controllers, peripherals, interfaces, oscillators, timing devices, video cards, audio cards, multimedia input/output (I/O) components, power supplies, and so forth. The embodiments, however, are not limited to implementation by the computing architecture **900**.

[0073] As shown in FIG. **9**, the computing architecture **900** comprises a processing unit **904**, a system memory **906** and a system bus **908**. The processing unit **904** (also referred to as a processor circuit) can be any of various commercially available processors, including without limitation an AMD® Athlon®, Duron® and Opteron® processors; ARM® application, embedded and secure processors; IBM® and Motorola® DragonBall® and PowerPC® processors; IBM and Sony® Cell processors; Intel® Celeron®, Core (2) Duo®, Itanium®, Pentium®, Xeon®, and XScale® processors; and similar processors. Dual microprocessors, multi-core processors, and other multi-processor architectures may also be employed as the processing unit **904**.

[0074] The system bus **908** provides an interface for system components including, but not limited to, the system memory **906** to the processing unit **904**. The system bus **908** can be any of several types of bus structure that may further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. Interface adapters may connect to the system bus **908** via a slot architecture. Example slot architectures may include without limitation Accelerated Graphics Port (AGP), Card Bus, (Extended) Industry Standard Architecture ((E)ISA), Micro Channel Architecture (MCA), NuBus, Peripheral Component Interconnect (Extended) (PCI(X)), PCI Express, Personal Computer Memory Card International Association (PCMCIA), and the like.

[0075] The system memory **906** may include various types of computer-readable storage media in the form of one or more higher speed memory units, such as read-only memory (ROM), random-access memory (RAM), dynamic RAM (DRAM), Double-Data-Rate DRAM (DDRAM), synchronous DRAM (SDRAM), bulk byte-addressable persistent memory (PMEM), static RAM (SRAM), programmable ROM (PROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), flash memory (e.g., one or more flash arrays), polymer memory such as ferroelectric polymer memory, ovonic memory, phase change or ferroelectric memory, silicon-oxide-nitride-oxide-silicon (SONOS) memory, magnetic or optical cards, an array of devices such as Redundant Array of Independent Disks (RAID) drives, solid state memory devices (e.g., USB memory, solid state drives (SSD) and any other type of storage media suitable for storing information. In the illustrated embodiment shown in FIG. **9**, the system memory **906** can include non-volatile memory **910** and/or volatile memory **912**. A basic input/output system (BIOS) can be stored in the non-volatile memory **910**.

[0076] The computer **902** may include various types of computer-readable storage media in the form of one or more lower speed memory units, including an internal (or external) hard disk drive (HDD) **914**, a magnetic floppy disk drive (FDD) **916** to read from or write to a removable magnetic disk **918**, and an optical disk drive **920** to read from or write to a removable optical disk **922** (e.g., a

compact disc read-only memory (CD-ROM) or digital versatile disc (DVD). The HDD **914**, FDD **916** and optical disk drive **920** can be connected to the system bus **908** by a HDD interface **924**, an FDD interface **926** and an optical drive interface **928**, respectively. The HDD interface **924** for external drive implementations can include at least one or both of Universal Serial Bus (USB) and IEEE 1394 interface technologies.

[0077] The drives and associated computer-readable media provide volatile and/or nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For example, a number of program modules can be stored in the drives and memory units **910**, **912**, including an operating system **930**, one or more application programs **932**, other program modules **934**, and program data **936**. In one embodiment, the one or more application programs **932**, other program modules **934**, and program data **936** can include, for example, the various applications and/or components of the system **100**, including the trajectory module **104**, the CV algorithms **106**, the social graph **107**, the stochastic model **108**, the decoder model **109**, the LSTMs **110**, **111**, the navigation logic **112**, the social graph network **113**, and/or other logic described herein.

[0078] A user can enter commands and information into the computer **902** through one or more wire/wireless input devices, for example, a keyboard **938** and a pointing device, such as a mouse **940**. Other input devices may include microphones, infra-red (IR) remote controls, radio-frequency (RF) remote controls, game pads, stylus pens, card readers, dongles, finger print readers, gloves, graphics tablets, joysticks, keyboards, retina readers, touch screens (e.g., capacitive, resistive, etc.), trackballs, trackpads, sensors, styluses, and the like. These and other input devices are often connected to the processing unit **904** through an input device interface **942** that is coupled to the system bus **908**, but can be connected by other interfaces such as a parallel port, IEEE 1394 serial port, a game port, a USB port, an IR interface, and so forth.

[0079] A monitor **944** or other type of display device is also connected to the system bus **908** via an interface, such as a video adaptor **946**. The monitor **944** may be internal or external to the computer **902**. In addition to the monitor **944**, a computer typically includes other peripheral output devices, such as speakers, printers, and so forth.

[0080] The computer **902** may operate in a networked environment using logical connections via wire and/or wireless communications to one or more remote computers, such as a remote computer **948**. In various embodiments, one or more migrations may occur via the networked environment. The remote computer **948** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer **902**, although, for purposes of brevity, only a memory/storage device **950** is illustrated. The logical connections depicted include wire/wireless connectivity to a local area network (LAN) **952** and/or larger networks, for example, a wide area network (WAN) **954**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which may connect to a global communications network, for example, the Internet.

[0081] When used in a LAN networking environment, the computer **902** is connected to the LAN **952** through a wire and/or wireless communication network interface or adaptor **956**. The adaptor **956** can facilitate wire and/or wireless communications to the LAN **952**, which may also include a wireless access point disposed thereon for communicating with the wireless functionality of the adaptor **956**.

[0082] When used in a WAN networking environment, the computer **902** can include a modem **958**, or is connected to a communications server on the WAN **954**, or has other means for establishing communications over the WAN **954**, such as by way of the Internet. The modem **958**, which can be internal or external and a wire and/or wireless device, connects to the system bus **908** via the input device interface **942**. In a networked environment, program modules depicted relative to the computer **902**, or portions thereof, can be stored in the remote memory/storage device **950**. It will be appreciated that the network connections shown are exemplary and other means of establishing a communications link between the computers can be used.

[0083] The computer **902** is operable to communicate with wire and wireless devices or entities using the IEEE 902 family of standards, such as wireless devices operatively disposed in wireless communication (e.g., IEEE 902.16 over-the-air modulation techniques). This includes at least Wi-Fi (or Wireless Fidelity), WiMax, and Bluetooth™ wireless technologies, among others. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices. Wi-Fi networks use radio technologies called IEEE 902.11x (a, b, g, n, ac, ay, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wire networks (which use IEEE 902.3-related media and functions).

[0084] One or more aspects of at least one example may be implemented by representative instructions stored on at least one machine-readable medium which represents various logic within the processor, which when read by a machine, computing device or system causes the machine, computing device or system to fabricate logic to perform the techniques described herein. Such representations, known as "IP cores" may be stored on a tangible, machine readable medium and supplied to various customers or manufacturing facilities to load into the fabrication machines that make the logic or processor.

[0085] Various examples may be implemented using hardware elements, software elements, or a combination of both. In some examples, hardware elements may include devices, components, processors, microprocessors, circuits, circuit elements (e.g., transistors, resistors, capacitors, inductors, and so forth), integrated circuits, application specific integrated circuits (ASIC), programmable logic devices (PLD), digital signal processors (DSP), field programmable gate array (FPGA), memory units, logic gates, registers, semiconductor device, chips, microchips, chip sets, and so forth. In some examples, software elements may include software components, programs, applications, computer programs, application programs, system programs, machine programs, operating system software, middleware, firmware, software modules, routines, subroutines, functions, methods, procedures, software interfaces, application program interfaces (API), instruction sets, computing code, computer code, code segments, computer code segments, words, values,

symbols, or any combination thereof. Determining whether an example is implemented using hardware elements and/or software elements may vary in accordance with any number of factors, such as desired computational rate, power levels, heat tolerances, processing cycle budget, input data rates, output data rates, memory resources, data bus speeds and other design or performance constraints, as desired for a given implementation.

[0086] Some examples may include an article of manufacture or at least one computer-readable medium. A computer-readable medium may include a non-transitory storage medium to store logic. In some examples, the non-transitory storage medium may include one or more types of computer-readable storage media capable of storing electronic data, including volatile memory or non-volatile memory, removable or non-removable memory, erasable or non-erasable memory, writeable or re-writeable memory, and so forth. In some examples, the logic may include various software elements, such as software components, programs, applications, computer programs, application programs, system programs, machine programs, operating system software, middleware, firmware, software modules, routines, subroutines, functions, methods, procedures, software interfaces, API, instruction sets, computing code, computer code, code segments, computer code segments, words, values, symbols, or any combination thereof.

[0087] According to some examples, a computer-readable medium may include a non-transitory storage medium to store or maintain instructions that when executed by a machine, computing device or system, cause the machine, computing device or system to perform methods and/or operations in accordance with the described examples. The instructions may include any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, and the like. The instructions may be implemented according to a predefined computer language, manner or syntax, for instructing a machine, computing device or system to perform a certain function. The instructions may be implemented using any suitable high-level, low-level, object-oriented, visual, compiled and/or interpreted programming language.

[0088] Some examples may be described using the expression "in one example" or "an example" along with their derivatives. These terms mean that a particular feature, structure, or characteristic described in connection with the example is included in at least one example. The appearances of the phrase "in one example" in various places in the specification are not necessarily all referring to the same example.

[0089] Some examples may be described using the expression "coupled" and "connected" along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, descriptions using the terms "connected" and/or "coupled" may indicate that two or more elements are in direct physical or electrical contact with each other. The term "coupled," however, may also mean that two or more elements are not in direct contact with each other, yet still co-operate or interact with each other.

[0090] The following examples pertain to further embodiments, from which numerous permutations and configurations will be apparent.

[0091] Example 1 is a non-transitory computer-readable storage medium storing instructions that when executed by a processor circuit cause the processor circuit to: determine a first feature vector describing destination features of a first person, of a plurality of people, depicted in an image; generate a directed graph for the image based on the plurality of people depicted in the image; determine, for the first person, a second feature vector based on the directed graph and the destination features; sample a value of a latent variable from a learned prior distribution, the latent variable to correspond to a first time interval; and generate, based on the sampled value and the feature vectors by a hierarchical long short-term memory (LSTM) comprising at least two LSTMs, an output vector comprising a direction of movement and a speed of the direction of movement of the first person at a second time interval, subsequent to the first time interval.

[0092] Example 2 includes the subject matter of example 1, the first feature vector to be determined by a first neural network based on input to comprise a location of the person at the first time interval and a velocity of the person at the first time interval, the velocity of the person at the first time interval based on the location of the person at the first time interval and a location of the first person at a third time interval, the third time interval prior to the first time interval, the first neural network to compute the first feature vector based at least in part on the following equation: $f_{j,t}^{D}=W^{d}[p_{j,t}, v_{j,t}]+b^{d}$.

[0093] Example 3 includes the subject matter of example 2, the location of the person at the first and third time intervals to comprise a respective set of polar coordinates, the output vector one of a plurality of output vectors generated for the first person, the output vectors to be computed based at least in part on the following equation: $p_{\theta}(v_t|z_{\leq t}, f_{<t}^{S}, f_{<t}^{D})=\mathrm{LSTM}_{\theta}(z_t, f_{t-1}^{S}, f_{t-1}^{D})$.

[0094] Example 4 includes the subject matter of example 2, the directed graph to comprise an adjacency matrix, a plurality of nodes, and a plurality of edges, each node to correspond to a respective one of the plurality of people depicted in the image, each edge connecting two of the nodes based on one of the plurality of people depicted in the image being in view of another one of the plurality of people depicted in the image, the adjacency matrix reflecting whether an edge connects two of the nodes.

[0095] Example 5 includes the subject matter of example 4, a first edge of the plurality of edges to connect a first node and a second node of the plurality of nodes, the computer-readable storage medium storing instructions that when executed by the processor circuit cause the processor circuit to: analyze the image to determine that the first person is in view of a second person of the plurality of people; generate the first edge in the graph, the first edge directed from the second node to the first node; determine an attention value for the first edge; and assign the attention value to the first edge.

[0096] Example 6 includes the subject matter of example 5, storing instructions that when executed by the processor circuit cause the processor circuit to: determine the second feature vector by the first neural network based on a message representing the first edge, the message to be determined based on an element-wise multiplication operator to be applied by the first neural network to the input to the first neural network, the element-wise operator to comprise a sigmoid activation function, the first neural network to determine the second feature vector based at least in part on one or more of the following equations:

$$f_j^{L+1} = f_j^L + \left(W^g \left(\sum_{\forall i, a_{ij}=1} M_{ij} (W^i x_{ij}^L + b^i) \right) + b^g \right);$$

$$x_{ij}^L = [f_i^L, f_j^L, W^r Polar_{pj}(p_i) + b^r)]; M_{ij}^L = \sum_{\forall i, a_{ij}=1} \alpha_{ij}^L \cdot (x_{ij}^L \odot g_{ij}^L);$$

$$\alpha_{ij}^l = \frac{\exp(Leaky\mathrm{Re}LU(W^a x_{ij}^L))}{\sum_{\forall i, a_{ij}=1} \exp(Leaky\mathrm{Re}LU(W^a x_{ij}^L))}; g_{ij}^L = \text{sigmoid } (W^s x_{ij}^L + b^s).$$

[0097] Example 7 includes the subject matter of example 1, the computer-readable storage medium storing instructions that when executed by the processor circuit cause the processor circuit to: learn the prior distribution based on a plurality of recursive hidden states of a posterior LSTM and a prior vector of the first person at a third time interval, the third time interval prior to the first time interval, the prior vector to comprise a direction of movement and a speed of the direction of movement of the first person at the second time interval, the prior distribution learned based at least in part on the following equation: $p_\psi(z_t|f_{<t}^S)=LSTM_\psi(f_{t-1}^S)$, the value for the latent variable sampled based at least in part on the following equation: $q_\phi(z_t|f_{\leq t}^S)=LSTM_\phi(f_t^S)$.

[0098] Example 8 includes the subject matter of example 1, the hierarchical LSTM comprising at least two LSTMs including a first LSTM and a second LSTM, the first LSTM to receive the value of the latent variable and the second feature vector as input, the second LSTM to receive an output of the first LSTM and the first feature vector as input, the second LSTM to generate the output vector.

[0099] Example 9 includes the subject matter of example 1, storing instructions that when executed by the processor circuit cause the processor circuit to: generate, for each person depicted in the image, a respective output vector comprising a direction of movement and a speed of the direction of movement of the respective person at the second time interval.

[0100] Example 10 includes the subject matter of example 1, storing instructions that when executed by the processor circuit cause the processor circuit to: receive a second image corresponding to a third time interval subsequent to the first and second time intervals; determine a third feature vector describing destination features of the first person depicted in the second image; generate a second directed graph for the second image based on a plurality of people depicted in the image; determine, for the first person, a fourth feature vector based on the second directed graph and the third feature vector; sample a second value for the latent variable from the learned prior distribution, the second value for the latent variable to correspond to the third time interval; and generate, based on the sampled second value and the third and fourth feature vectors by the hierarchical LSTM, a second output vector comprising a direction of movement and a speed of the direction of movement of the first person at the third time interval.

[0101] Example 11 includes the subject matter of example 1, storing instructions that when executed by the processor circuit cause the processor circuit to: determine that the output vector intersects with a trajectory vector of an autonomous vehicle; determine, based on the determined intersection of the output vector and the trajectory vector, that a likelihood of a collision between the first person and an autonomous vehicle at a later time exceeds a collision threshold; and output an indication of the collision to at least one of the first person or the autonomous vehicle, a navigation logic of the autonomous vehicle to modify a movement of the autonomous vehicle to avoid the collision responsive to receipt of the indication.

[0102] Example 12 includes an apparatus, comprising: a processor circuit; and memory storing instructions which when executed by the processor circuit cause the processor circuit to: determine a first feature vector describing destination features of a first person, of a plurality of people, depicted in an image; generate a directed graph for the image based on the plurality of people depicted in the image; determine, for the first person, a second feature vector based on the directed graph and the destination features; sample a value of a latent variable from a learned prior distribution, the latent variable to correspond to a first time interval; and generate, based on the sampled value and the feature vectors by a hierarchical long short-term memory (LSTM) comprising at least two LSTMs, an output vector comprising a direction of movement and a speed of the direction of movement of the first person at a second time interval, subsequent to the first time interval.

[0103] Example 13 includes the subject matter of example 12, the first feature vector to be determined by a first neural network based on input to comprise a location of the person at the first time interval and a velocity of the person at the first time interval, the velocity of the person at the first time interval based on the location of the person at the first time interval and a location of the first person at a third time interval, the third time interval prior to the first time interval, the first neural network to compute the first feature vector based at least in part on the following equation: $f_{j,t}^D=W^d[p_{j,t}, v_{j,t}]+b^d$.

[0104] Example 14 includes the subject matter of example 13, the location of the person at the first and third time intervals to comprise a respective set of polar coordinates, the output vector one of a plurality of output vectors generated for the first person, the output vectors to be computed based at least in part on the following equation: $p_\theta(v_t|z_{\leq t}, f_{<t}, f_{<t}^D)=LSTM_\theta(z_t, f_{t-1}^S, f_{t-1}^D)$ . . . .

[0105] Example 15 includes the subject matter of example 13, the directed graph to comprise an adjacency matrix, a plurality of nodes, and a plurality of edges, each node to correspond to a respective one of the plurality of people depicted in the image, each edge connecting two of the nodes based on one of the plurality of people depicted in the image being in view of another one of the plurality of people depicted in the image, the adjacency matrix reflecting whether an edge connects two of the nodes.

[0106] Example 16 includes the subject matter of example 15, a first edge of the plurality of edges to connect a first node and a second node of the plurality of nodes, the memory storing instructions that when executed by the processor circuit cause the processor circuit to: analyze the image to determine that the first person is in view of a second person of the plurality of people; generate the first edge in the graph, the first edge directed from the second node to the first node; determine an attention value for the first edge; and assign the attention value to the first edge.

[0107] Example 17 includes the subject matter of example 16, the memory storing instructions which when executed by the processor circuit cause the processor circuit to: determine the second feature vector by the first neural network based on a message representing the first edge, the message to be determined based on an element-wise multiplication operator to be applied by the first neural network

to the input to the first neural network, the element-wise operator to comprise a sigmoid activation function, the first neural network to determine the second feature vector based at least in part on one or more of the following equations:

$$f_j^{L+1} = f_j^L + \left(W^g\left(\sum_{\forall i, a_{ij}=1} M_{ij}(W^i x_{ij}^L + b^i)\right) + b^g\right);$$

$$x_{ij}^L = [f_i^L, f_j^L, W^r Polar_{pj}(p_i) + b^r)]; M_{ij}^L = \sum_{\forall i, a_{ij}=1} \alpha_{ij}^L \cdot (x_{ij}^L \odot g_{ij}^L);$$

$$\alpha_{ij}^l = \frac{\exp(Leaky\mathrm{Re}LU(W^a x_{ij}^L))}{\sum_{\forall i, a_{ij}=1} \exp(Leaky\mathrm{Re}LU(W^a x_{ij}^L))}; g_{ij}^L = \text{sigmoid } (W^s x_{ij}^L + b^s).$$

[0108] Example 18 includes the subject matter of example 12, the memory storing instructions which when executed by the processor circuit cause the processor circuit to: learn the prior distribution based on a plurality of recursive hidden states of a posterior LSTM and a prior vector of the first person at a third time interval, the third time interval prior to the first time interval, the prior vector to comprise a direction of movement and a speed of the direction of movement of the first person at the second time interval, the prior distribution learned based at least in part on the following equation: $p_\psi(z_t|f_{<t}^S)=LSTM_\psi(f_{t-1}^S)$, the value for the latent variable sampled based at least in part on the following equation: $q_\phi(z_t|f_{\leq t}^S)=LSTM_\phi(f_t^S)$.

[0109] Example 19 includes the subject matter of example 12, the hierarchical LSTM comprising at least two LSTMs including a first LSTM and a second LSTM, the first LSTM to receive the value of the latent variable and the second feature vector as input, the second LSTM to receive an output of the first LSTM and the first feature vector as input, the second LSTM to generate the output vector.

[0110] Example 20 includes the subject matter of example 12, the memory storing instructions which when executed by the processor circuit cause the processor circuit to: generate, for each person depicted in the image, a respective output vector comprising a direction of movement and a speed of the direction of movement of the respective person at the second time interval.

[0111] Example 21 includes the subject matter of example 12, the memory storing instructions which when executed by the processor circuit cause the processor circuit to: receive a second image corresponding to a third time interval subsequent to the first and second time intervals; determine a third feature vector describing destination features of the first person depicted in the second image; generate a second directed graph for the second image based on a plurality of people depicted in the image; determine, for the first person, a fourth feature vector based on the second directed graph and the third feature vector; sample a second value for the latent variable from the learned prior distribution, the second value for the latent variable to correspond to the third time interval; and generate, based on the sampled second value and the third and fourth feature vectors by the hierarchical LSTM, a second output vector comprising a direction of movement and a speed of the direction of movement of the first person at the third time interval.

[0112] Example 22 includes the subject matter of example 12, the memory storing instructions which when executed by the processor circuit cause the processor circuit to: determine that the output vector intersects with a trajectory vector of an autonomous vehicle; determining, based on the determined intersection of the output vector and the trajectory vector, that a likelihood of a collision between the first person and an autonomous vehicle at a later time exceeds a collision threshold; and output an indication of the collision to at least one of the first person or the autonomous vehicle, a navigation logic of the autonomous vehicle to modify a movement of the autonomous vehicle to avoid the collision responsive to receipt of the indication.

[0113] Example 23 includes a method, comprising: determining a first feature vector describing destination features of a first person, of a plurality of people, depicted in an image; generating a directed graph for the image based on the plurality of people depicted in the image; determining, for the first person, a second feature vector based on the directed graph and the destination features; sampling a value of a latent variable from a learned prior distribution, the latent variable to correspond to a first time interval; and generating, based on the sampled value and the feature vectors by a hierarchical long short-term memory (LSTM) executing on a processor, an output vector comprising a direction of movement and a speed of the direction of movement of the first person at a second time interval, subsequent to the first time interval.

[0114] Example 24 includes the subject matter of example 23, the first feature vector to be determined by a first neural network based on input to comprise a location of the person at the first time interval and a velocity of the person at the first time interval, the velocity of the person at the first time interval based on the location of the person at the first time interval and a location of the first person at a third time interval, the third time interval prior to the first time interval, the first neural network to compute the first feature vector based at least in part on the following equation: $f_{j,t}^D=W^d[p_{j,t}, v_{j,t}]+b^d$.

[0115] Example 25 includes the subject matter of example 24, the location of the person at the first and third time intervals to comprise a respective set of polar coordinates, the output vector one of a plurality of output vectors generated for the first person, the output vectors to be computed based at least in part on the following equation: $p_\theta(v_t|z_{\leq t}, f_{<t}^S, f_{<t}^D)=LSTM_\theta(z_t, f_{t-1}^S, f_{t-1}^D)$.

[0116] Example 26 includes the subject matter of example 24, the directed graph to comprise an adjacency matrix, a plurality of nodes, and a plurality of edges, each node to correspond to a respective one of the plurality of people depicted in the image, each edge connecting two of the nodes based on one of the plurality of people depicted in the image being in view of another one of the plurality of people depicted in the image, the adjacency matrix reflecting whether an edge connects two of the nodes.

[0117] Example 27 includes the subject matter of example 26, a first edge of the plurality of edges to connect a first node and a second node of the plurality of nodes, the method further comprising: analyzing the image to determine that the first person is in view of a second person of the plurality of people; generating the first edge in the graph, the first edge directed from the second node to the first node; determining an attention value for the first edge; and assigning the attention value to the first edge.

[0118] Example 28 includes the subject matter of example 27, further comprising: determining the second feature vector by the first neural network based on a message representing the first edge, the message to be determined based on an element-wise multiplication operator to be applied by the

first neural network to the input to the first neural network, the element-wise operator to comprise a sigmoid activation function, the first neural network to determine the second feature vector based at least in part on one or more of the following equations:

$$f_j^{L+1} = f_j^L + \left(W^g \left(\sum_{\forall i, a_{ij}=1} M_{ij} (W^i x_{ij}^L + b^i) \right) + b^g \right);$$

$$x_{ij}^L = [f_i^L, f_j^L, W^r Polar_{pj}(p_i) + b^r)]; M_{ij}^L = \sum_{\forall i, a_{ij}=1} \alpha_{ij}^L \cdot (x_{ij}^L \odot g_{ij}^L);$$

$$\alpha_{ij}^l = \frac{\exp(Leaky\mathrm{Re}LU(W^a x_{ij}^L))}{\sum_{\forall i, a_{ij}=1} \exp(Leaky\mathrm{Re}LU(W^a x_{ij}^L))}; g_{ij}^L = \text{sigmoid } (W^s x_{ij}^L + b^s).$$

[0119] Example 29 includes the subject matter of example 23, further comprising: learning the prior distribution based on a plurality of recursive hidden states of a posterior LSTM and a prior vector of the first person at a third time interval, the third time interval prior to the first time interval, the prior vector to comprise a direction of movement and a speed of the direction of movement of the first person at the second time interval, the prior distribution learned based at least in part on the following equation: $p_\psi(z_t|f_{<t}^S)=LSTM_\psi(f_{t-1}^S)$, the value for the latent variable sampled based at least in part on the following equation: $q_\phi(z_t|f_{\leq t}^S)=LSTM_\phi(f_t^S)$.

[0120] Example 30 includes the subject matter of example 23, the hierarchical LSTM comprising at least two LSTMs including a first LSTM and a second LSTM, the first LSTM to receive the value of the latent variable and the second feature vector as input, the second LSTM to receive an output of the first LSTM and the first feature vector as input, the second LSTM to generate the output vector.

[0121] Example 31 includes the subject matter of example 23, further comprising: generating, for each person depicted in the image, a respective output vector comprising a direction of movement and a speed of the direction of movement of the respective person at the second time interval.

[0122] Example 32 includes the subject matter of example 23, further comprising: receiving a second image corresponding to a third time interval subsequent to the first and second time intervals; determining a third feature vector describing destination features of the first person depicted in the second image; generating a second directed graph for the second image based on a plurality of people depicted in the image; determining, for the first person, a fourth feature vector based on the second directed graph and the third feature vector; sampling a second value for the latent variable from the learned prior distribution, the second value for the latent variable to correspond to the third time interval; and generating, based on the sampled second value and the third and fourth feature vectors by the hierarchical LSTM, a second output vector comprising a direction of movement and a speed of the direction of movement of the first person at the third time interval.

[0123] Example 33 includes the subject matter of example 23, further comprising: determining that the output vector intersects with a trajectory vector of an autonomous vehicle; determining, based on the determined intersection of the output vector and the trajectory vector, that a likelihood of a collision between the first person and an autonomous vehicle at a later time exceeds a collision threshold; outputting an indication of the collision to at least one of the first person or the autonomous vehicle; receiving, by a navigation logic of the autonomous vehicle, the indication of the collision; and modifying, by the navigation logic, a movement of the autonomous vehicle to avoid the collision.

[0124] Example 34 is an apparatus, comprising: means for determining a first feature vector describing destination features of a first person, of a plurality of people, depicted in an image; means for generating a directed graph for the image based on the plurality of people depicted in the image; means for determining, for the first person, a second feature vector based on the directed graph and the destination features; means for sampling a value of a latent variable from a learned prior distribution, the latent variable to correspond to a first time interval; and means for generating, based on the sampled value and the feature vectors by a hierarchical long short-term memory (LSTM) comprising at least two LSTMs, an output vector comprising a direction of movement and a speed of the direction of movement of the first person at a second time interval, subsequent to the first time interval.

[0125] Example 35 includes the subject matter of example 34, the first feature vector to be determined by a first neural network based on input to comprise a location of the person at the first time interval and a velocity of the person at the first time interval, the velocity of the person at the first time interval based on the location of the person at the first time interval and a location of the first person at a third time interval, the third time interval prior to the first time interval, the first neural network to compute the first feature vector based at least in part on the following equation: $f_{j,t}^D=W^d[p_{j,t}, v_{j,t}]+b^d$.

[0126] Example 36 includes the subject matter of example 35, the location of the person at the first and third time intervals to comprise a respective set of polar coordinates, the output vector one of a plurality of output vectors generated for the first person, the output vectors to be computed based at least in part on the following equation: $p_\theta(v_t|z_{\leq t}, f_{<t}^S, f_{<t}^D)=LSTM_\theta(z_t, f_{t-1}^S, f_{t-1}^D)$.

[0127] Example 37 includes the subject matter of example 35, the directed graph to comprise an adjacency matrix, a plurality of nodes, and a plurality of edges, each node to correspond to a respective one of the plurality of people depicted in the image, each edge connecting two of the nodes based on one of the plurality of people depicted in the image being in view of another one of the plurality of people depicted in the image, the adjacency matrix reflecting whether an edge connects two of the nodes.

[0128] Example 38 includes the subject matter of example 37, a first edge of the plurality of edges to connect a first node and a second node of the plurality of nodes, the apparatus further comprising: means for analyzing the image to determine that the first person is in view of a second person of the plurality of people; generating the first edge in the graph, the first edge directed from the second node to the first node; means for determining an attention value for the first edge; and means for assigning the attention value to the first edge.

[0129] Example 39 includes the subject matter of example 38, further comprising: means for determining the second feature vector by the first neural network based on a message representing the first edge, the message to be determined based on an element-wise multiplication operator to be applied by the first neural network to the input to the first neural network, the element-wise operator to comprise a sigmoid activation function, the first neural network to

determine the second feature vector based at least in part on one or more of the following equations:

$$f_j^{L+1} = f_j^L + \left(W^g\left(\sum_{\forall i, a_{ij}=1} M_{ij}(W^i x_{ij}^L + b^i)\right) + b^g\right);$$

$$x_{ij}^L = [f_i^L, f_j^L, W^r Polar_{pj}(p_i) + b^r)]; M_{ij}^L = \sum_{\forall i, a_{ij}=1} \alpha_{ij}^L \cdot (x_{ij}^L \odot g_{ij}^L);$$

$$\alpha_{ij}^l = \frac{\exp(Leaky\mathrm{ReLU}(W^a x_{ij}^L))}{\sum_{\forall i, a_{ij}=1} \exp(Leaky\mathrm{ReLU}(W^a x_{ij}^L))}; g_{ij}^L = \text{sigmoid } (W^s x_{ij}^L + b^s).$$

[0130] Example 40 includes the subject matter of example 34, further comprising: means for learning the prior distribution based on a plurality of recursive hidden states of a posterior LSTM and a prior vector of the first person at a third time interval, the third time interval prior to the first time interval, the prior vector to comprise a direction of movement and a speed of the direction of movement of the first person at the second time interval, the prior distribution learned based at least in part on the following equation: $p_\psi(z_t|f_{<t}^S)=LSTM_\psi(f_{t-1}^S)$, the value for the latent variable sampled based at least in part on the following equation: $q_\phi(z^t|f_{\leq t}^S)=LSTM_\phi(f_t^S)$.

[0131] Example 41 includes the subject matter of example 34, the hierarchical LSTM comprising at least two LSTMs including a first LSTM and a second LSTM, the first LSTM to receive the value of the latent variable and the second feature vector as input, the second LSTM to receive an output of the first LSTM and the first feature vector as input, the second LSTM to generate the output vector.

[0132] Example 42 includes the subject matter of example 34, further comprising: means for generating, for each person depicted in the image, a respective output vector comprising a direction of movement and a speed of the direction of movement of the respective person at the second time interval.

[0133] Example 43 includes the subject matter of example 34, further comprising: means for receiving a second image corresponding to a third time interval subsequent to the first and second time intervals; means for determining a third feature vector describing destination features of the first person depicted in the second image; means for generating a second directed graph for the second image based on a plurality of people depicted in the image; means for determining, for the first person, a fourth feature vector based on the second directed graph and the third feature vector; means for sampling a second value for the latent variable from the learned prior distribution, the second value for the latent variable to correspond to the third time interval; and means for generating, based on the sampled second value and the third and fourth feature vectors by the hierarchical LSTM, a second output vector comprising a direction of movement and a speed of the direction of movement of the first person at the third time interval.

[0134] Example 44 includes the subject matter of example 34, further comprising: means for determining that the output vector intersects with a trajectory vector of an autonomous vehicle; means for determining, based on the determined intersection of the output vector and the trajectory vector, that a likelihood of a collision between the first person and an autonomous vehicle at a later time exceeds a collision threshold; means for outputting an indication of the collision to at least one of the first person or the autonomous vehicle; means for receiving, by a navigation logic of the autonomous vehicle, the indication of the collision; and means for modifying, by the navigation logic, a movement of the autonomous vehicle to avoid the collision.

[0135] In addition, in the foregoing, various features are grouped together in a single example to streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate example. In the appended claims, the terms "including" and "in which" are used as the plain-English equivalents of the respective terms "comprising" and "wherein," respectively. Moreover, the terms "first," "second," "third," and so forth, are used merely as labels, and are not intended to impose numerical requirements on their objects.

[0136] Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

[0137] A data processing system suitable for storing and/or executing program code will include at least one processor coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories which provide temporary storage of at least some program code to reduce the number of times code must be retrieved from bulk storage during execution. The term "code" covers a broad range of software components and constructs, including applications, drivers, processes, routines, methods, modules, firmware, microcode, and subprograms Thus, the term "code" may be used to refer to any collection of instructions which, when executed by a processing system, perform a desired operation or operations.

[0138] Logic circuitry, devices, and interfaces herein described may perform functions implemented in hardware and implemented with code executed on one or more processors. Logic circuitry refers to the hardware or the hardware and code that implements one or more logical functions. Circuitry is hardware and may refer to one or more circuits. Each circuit may perform a particular function. A circuit of the circuitry may comprise discrete electrical components interconnected with one or more conductors, an integrated circuit, a chip package, a chip set, memory, or the like. Integrated circuits include circuits created on a substrate such as a silicon wafer and may comprise components. And integrated circuits, processor packages, chip packages, and chipsets may comprise one or more processors.

[0139] Processors may receive signals such as instructions and/or data at the input(s) and process the signals to generate the at least one output. While executing code, the code changes the physical states and characteristics of transistors that make up a processor pipeline. The physical states of the transistors translate into logical bits of ones and zeros stored in registers within the processor. The processor can transfer

the physical states of the transistors into registers and transfer the physical states of the transistors to another storage medium.

[0140] A processor may comprise circuits to perform one or more sub-functions implemented to perform the overall function of the processor. One example of a processor is a state machine or an application-specific integrated circuit (ASIC) that includes at least one input and at least one output. A state machine may manipulate the at least one input to generate the at least one output by performing a predetermined series of serial and/or parallel manipulations or transformations on the at least one input.

[0141] The logic as described above may be part of the design for an integrated circuit chip. The chip design is created in a graphical computer programming language, and stored in a computer storage medium or data storage medium (such as a disk, tape, physical hard drive, or virtual hard drive such as in a storage access network). If the designer does not fabricate chips or the photolithographic masks used to fabricate chips, the designer transmits the resulting design by physical means (e.g., by providing a copy of the storage medium storing the design) or electronically (e.g., through the Internet) to such entities, directly or indirectly. The stored design is then converted into the appropriate format (e.g., GDSII) for the fabrication.

[0142] The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case, the chip is mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other higher level carrier) or in a multichip package (such as a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case, the chip is then integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either (a) an intermediate product, such as a processor board, a server platform, or a motherboard, or (b) an end product.

[0143] The foregoing description of example embodiments has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed. Many modifications and variations are possible in light of this disclosure. It is intended that the scope of the present disclosure be limited not by this detailed description, but rather by the claims appended hereto. Future filed applications claiming priority to this application may claim the disclosed subject matter in a different manner, and may generally include any set of one or more limitations as variously disclosed or otherwise demonstrated herein.

1-20. (canceled)

21. A non-transitory computer-readable storage medium storing instructions that when executed by a processor circuit cause the processor circuit to:

determine a first feature vector describing destination features of a first person, of a plurality of people, depicted in an image;

generate a directed graph for the image based on the plurality of people depicted in the image;

determine, for the first person, a second feature vector based on the directed graph and the destination features;

sample a value of a latent variable from a learned prior distribution, the latent variable to correspond to a first time interval; and

generate, based on the sampled value and the feature vectors by a hierarchical long short-term memory (LSTM), an output vector comprising a direction of movement and a speed of the direction of movement of the first person at a second time interval, subsequent to the first time interval.

22. The computer-readable storage medium of claim **21**, the first feature vector to be determined by a first neural network based on input to comprise a location of the person at the first time interval and a velocity of the person at the first time interval, the velocity of the person at the first time interval based on the location of the person at the first time interval and a location of the first person at a third time interval, the third time interval prior to the first time interval.

23. The computer-readable storage medium of claim **22**, the location of the person at the first and third time intervals to comprise a respective set of polar coordinates, the output vector one of a plurality of output vectors generated for the first person.

24. The computer-readable storage medium of claim **22**, the directed graph to comprise an adjacency matrix, a plurality of nodes, and a plurality of edges, each node to correspond to a respective one of the plurality of people depicted in the image, each edge connecting two of the nodes based on one of the plurality of people depicted in the image being in view of another one of the plurality of people depicted in the image, the adjacency matrix reflecting whether an edge connects two of the nodes.

25. The computer-readable storage medium of claim **24**, a first edge of the plurality of edges to connect a first node and a second node of the plurality of nodes, the computer-readable storage medium storing instructions that when executed by the processor circuit cause the processor circuit to:

analyze the image to identify an interaction from the first person directed to a second person of the plurality of people;

generate the first edge in the graph based on the identified interaction, the first person associated with the first node, the second person associated with the second node;

determine an attention value for the first edge, the attention value to reflect a degree of the interaction; and

assign the attention value to the first edge.

26. The computer-readable storage medium of claim **25**, storing instructions that when executed by the processor circuit cause the processor circuit to:

determine the second feature vector by the first neural network based on a message representing the first edge, the message to be determined based on an element-wise multiplication operator to be applied by the first neural network to the input to the first neural network, the element-wise operator to comprise a sigmoid activation function.

27. The computer-readable storage medium of claim **21**, the computer-readable storage medium storing instructions that when executed by the processor circuit cause the processor circuit to:

learn the prior distribution based on a plurality of recursive hidden states of a third LSTM and a prior vector of the first person at a third time interval, the third time

interval prior to the first time interval, the prior vector to comprise a direction of movement and a speed of the direction of movement of the first person at the second time interval.

28. The computer-readable storage medium of claim **21**, the hierarchical LSTM comprising at least two LSTMs including a first LSTM and a second LSTM, the first LSTM to receive the value of the latent variable and the second feature vector as input, the second LSTM to receive an output of the first LSTM and the first feature vector as input, the second LSTM to generate the output vector.

29. The computer-readable storage medium of claim **21**, storing instructions that when executed by the processor circuit cause the processor circuit to:

determine that the output vector intersects with a trajectory vector of an autonomous vehicle;

determine, based on the determined intersection of the output vector and the trajectory vector, that a likelihood of a collision between the first person and an autonomous vehicle at a later time exceeds a collision threshold; and

output an indication of the collision to at least one of the first person or the autonomous vehicle, a navigation logic of the autonomous vehicle to modify a movement of the autonomous vehicle to avoid the collision responsive to receipt of the indication.

30. An apparatus, comprising:

a processor circuit; and

memory storing instructions which when executed by the processor circuit cause the processor circuit to:

determine a first feature vector describing destination features of a first person, of a plurality of people, depicted in an image;

generate a directed graph for the image based on the plurality of people depicted in the image;

determine, for the first person, a second feature vector based on the directed graph and the destination features;

sample a value of a latent variable from a learned prior distribution, the latent variable to correspond to a first time interval; and

generate, based on the sampled value and the feature vectors by a hierarchical long short-term memory (LSTM), an output vector comprising a direction of movement and a speed of the direction of movement of the first person at a second time interval, subsequent to the first time interval.

31. The apparatus of claim **30**, the first feature vector to be determined by a first neural network based on input to comprise a location of the person at the first time interval and a velocity of the person at the first time interval, the velocity of the person at the first time interval based on the location of the person at the first time interval and a location of the first person at a third time interval, the third time interval prior to the first time interval.

32. The apparatus of claim **31**, the location of the person at the first and third time intervals to comprise a respective set of polar coordinates, the output vector one of a plurality of output vectors generated for the first person.

33. The apparatus of claim **31**, the directed graph to comprise an adjacency matrix, a plurality of nodes, and a plurality of edges, each node to correspond to a respective one of the plurality of people depicted in the image, each edge connecting two of the nodes based on one of the plurality of people depicted in the image being in view of another one of the plurality of people depicted in the image, the adjacency matrix reflecting whether an edge connects two of the nodes.

34. The apparatus of claim **33**, a first edge of the plurality of edges to connect a first node and a second node of the plurality of nodes, the memory storing instructions that when executed by the processor circuit cause the processor circuit to:

analyze the image to identify an interaction from the first person directed to a second person of the plurality of people;

generate the first edge in the graph based on the identified interaction, the first person associated with the first node, the second person associated with the second node;

determine an attention value for the first edge, the attention value to reflect a degree of the interaction; and

assign the attention value to the first edge.

35. The apparatus of claim **30**, the memory storing instructions which when executed by the processor circuit cause the processor circuit to:

learn the prior distribution based on a plurality of recursive hidden states of a third LSTM and a prior vector of the first person at a third time interval, the third time interval prior to the first time interval, the prior vector to comprise a direction of movement and a speed of the direction of movement of the first person at the second time interval.

36. The apparatus of claim **30**, the hierarchical LSTM comprising at least two LSTMs including a first LSTM and a second LSTM, the first LSTM to receive the value of the latent variable and the second feature vector as input, the second LSTM to receive an output of the first LSTM and the first feature vector as input, the second LSTM to generate the output vector.

37. The apparatus of claim **30**, the memory storing instructions which when executed by the processor circuit cause the processor circuit to:

determine that the output vector intersects with a trajectory vector of an autonomous vehicle;

determine, based on the determined intersection of the output vector and the trajectory vector, that a likelihood of a collision between the first person and an autonomous vehicle at a later time exceeds a collision threshold; and

output an indication of the collision to at least one of the first person or the autonomous vehicle, a navigation logic of the autonomous vehicle to modify a movement of the autonomous vehicle to avoid the collision responsive to receipt of the indication.

38. A method, comprising:

determining a first feature vector describing destination features of a first person, of a plurality of people, depicted in an image;

generating a directed graph for the image based on the plurality of people depicted in the image;

determining, for the first person, a second feature vector based on the directed graph and the destination features;

sampling a value of a latent variable from a learned prior distribution, the latent variable to correspond to a first time interval; and

generating, based on the sampled value and the feature vectors by a hierarchical long short-term memory (LSTM) executing on a processor, an output vector comprising a direction of movement and a speed of the direction of movement of the first person at a second time interval, subsequent to the first time interval.

39. The method of claim **38**, the first feature vector to be determined by a first neural network based on input to comprise a location of the person at the first time interval and a velocity of the person at the first time interval, the velocity of the person at the first time interval based on the location of the person at the first time interval and a location of the first person at a third time interval, the third time interval prior to the first time interval.

40. The method of claim **39**, the directed graph to comprise an adjacency matrix, a plurality of nodes, and a plurality of edges, each node to correspond to a respective one of the plurality of people depicted in the image, each edge connecting two of the nodes based on one of the plurality of people depicted in the image being in view of another one of the plurality of people depicted in the image, the adjacency matrix reflecting whether an edge connects two of the nodes.

41. The method of claim **40**, a first edge of the plurality of edges to connect a first node and a second node of the plurality of nodes, the method further comprising:

analyzing the image to identify an interaction from the first person directed to a second person of the plurality of people;

generating the first edge in the graph based on the identified interaction, the first person associated with the first node, the second person associated with the second node;

determining an attention value for the first edge, the attention value to reflect a degree of the interaction; and

assigning the attention value to the first edge.

42. The method of claim **41**, further comprising:

determining the second feature vector by the first neural network based on a message representing the first edge, the message to be determined based on an element-wise multiplication operator to be applied by the first neural network to the input to the first neural network, the element-wise operator to comprise a sigmoid activation function.

43. The method of claim **38**, further comprising:

learning the prior distribution based on a plurality of recursive hidden states of a third LSTM and a prior vector of the first person at a third time interval, the third time interval prior to the first time interval, the prior vector to comprise a direction of movement and a speed of the direction of movement of the first person at the second time interval.

44. The method of claim **38**, the hierarchical LSTM comprising at least two LSTMs including a first LSTM and a second LSTM, the first LSTM to receive the value of the latent variable and the second feature vector as input, the second LSTM to receive an output of the first LSTM and the first feature vector as input, the second LSTM to generate the output vector.

45. The method of claim **38**, further comprising:

determining that the output vector intersects with a trajectory vector of an autonomous vehicle;

determining, based on the determined intersection of the output vector and the trajectory vector, that a likelihood of a collision between the first person and an autonomous vehicle at a later time exceeds a collision threshold;

outputting an indication of the collision to at least one of the first person or the autonomous vehicle;

receiving, by a navigation logic of the autonomous vehicle, the indication of the collision; and

modifying, by the navigation logic, a movement of the autonomous vehicle to avoid the collision.

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